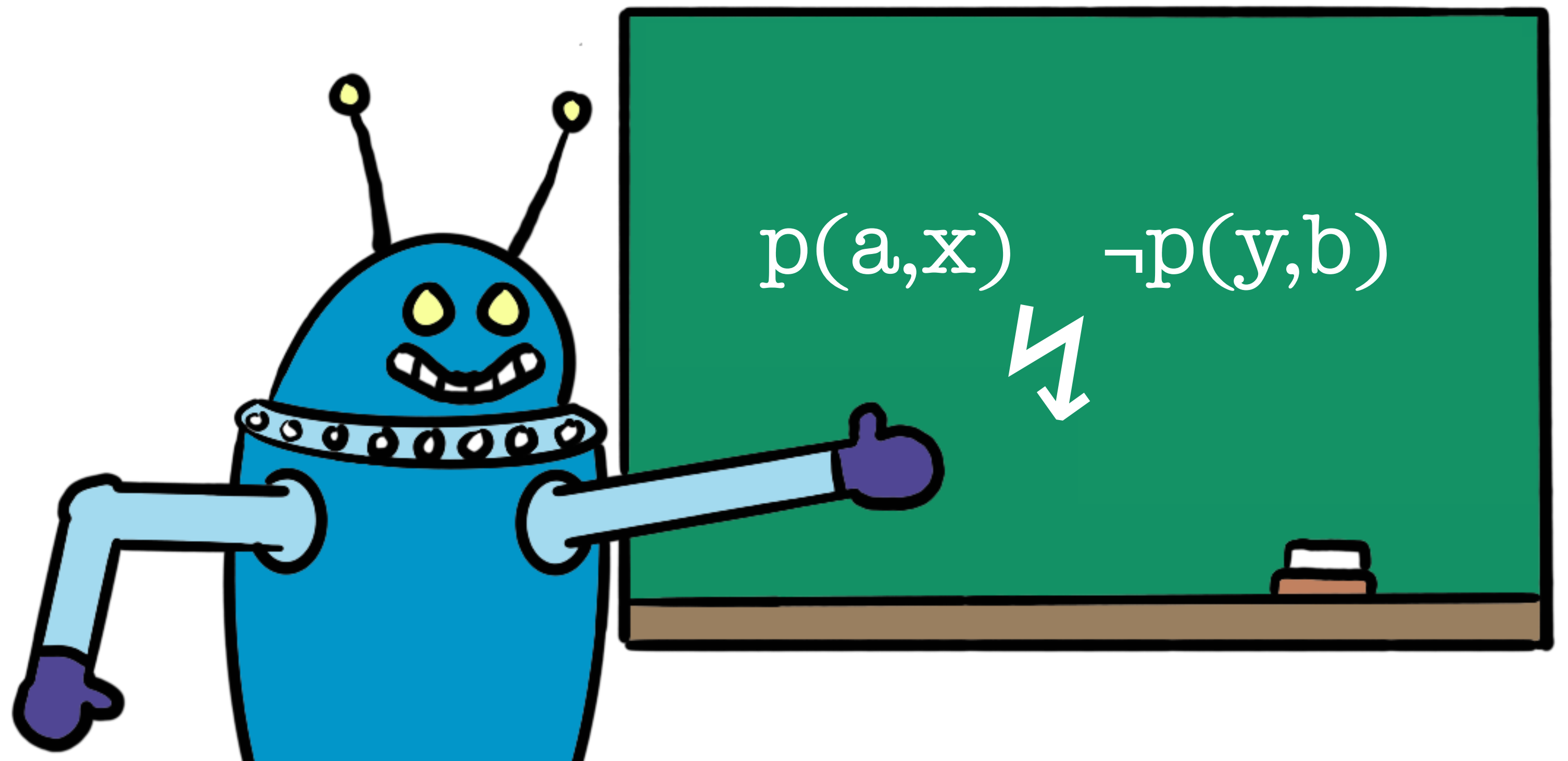


Provers and Solvers

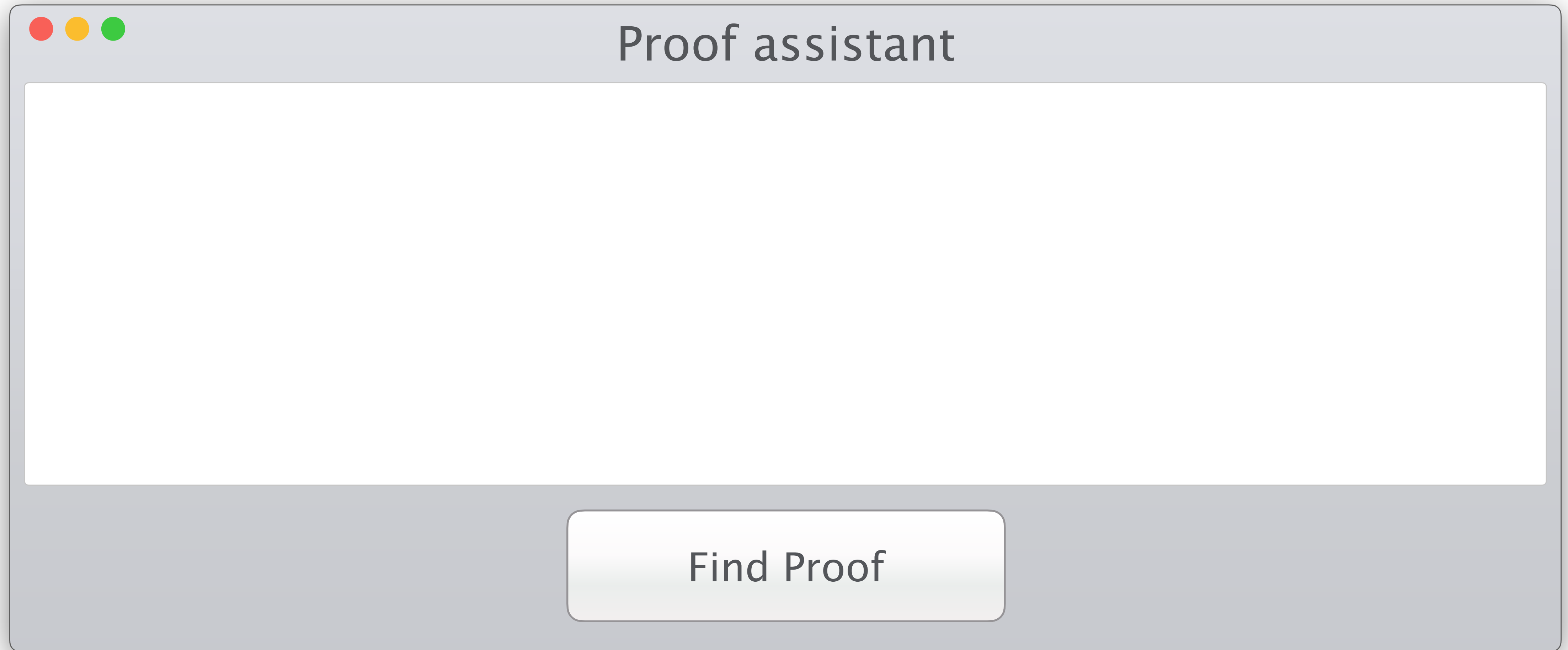
Lecture 1: Resolution

Jasmin Blanchette

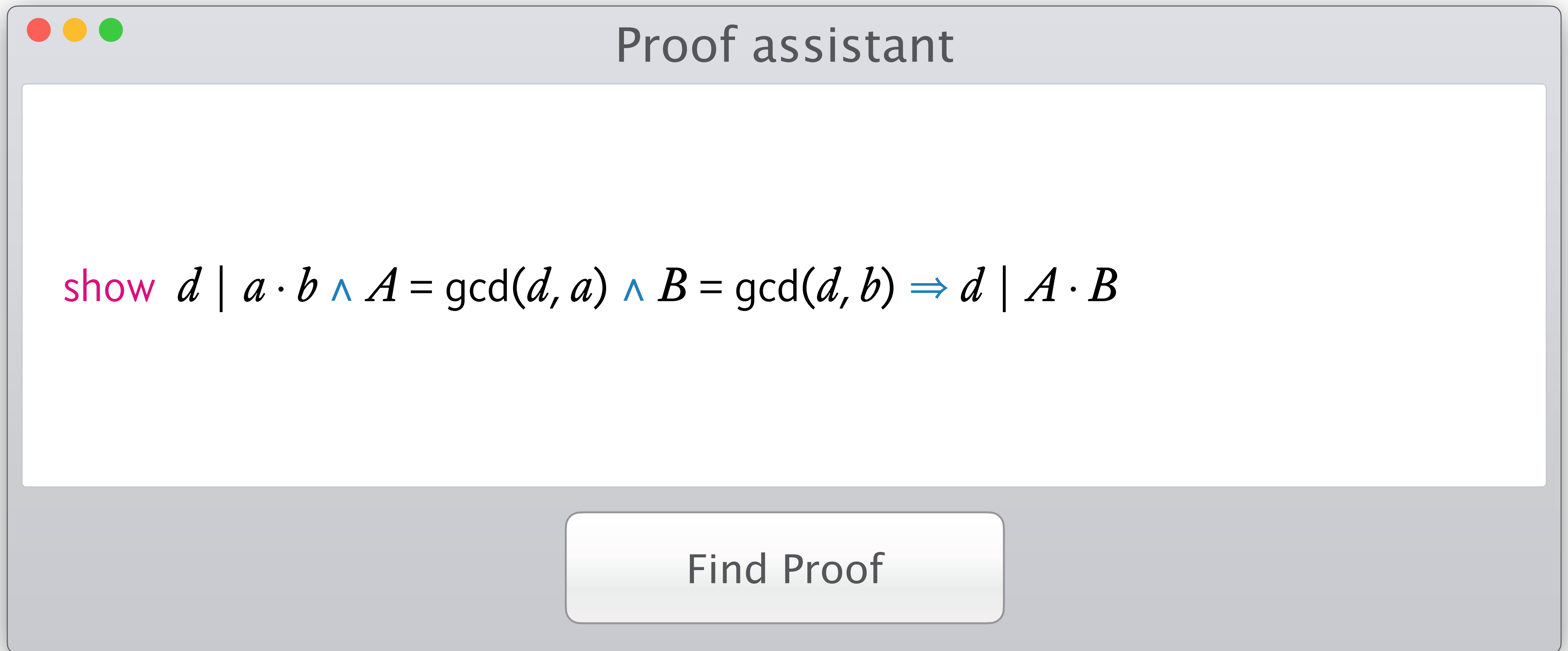
LMU Munich



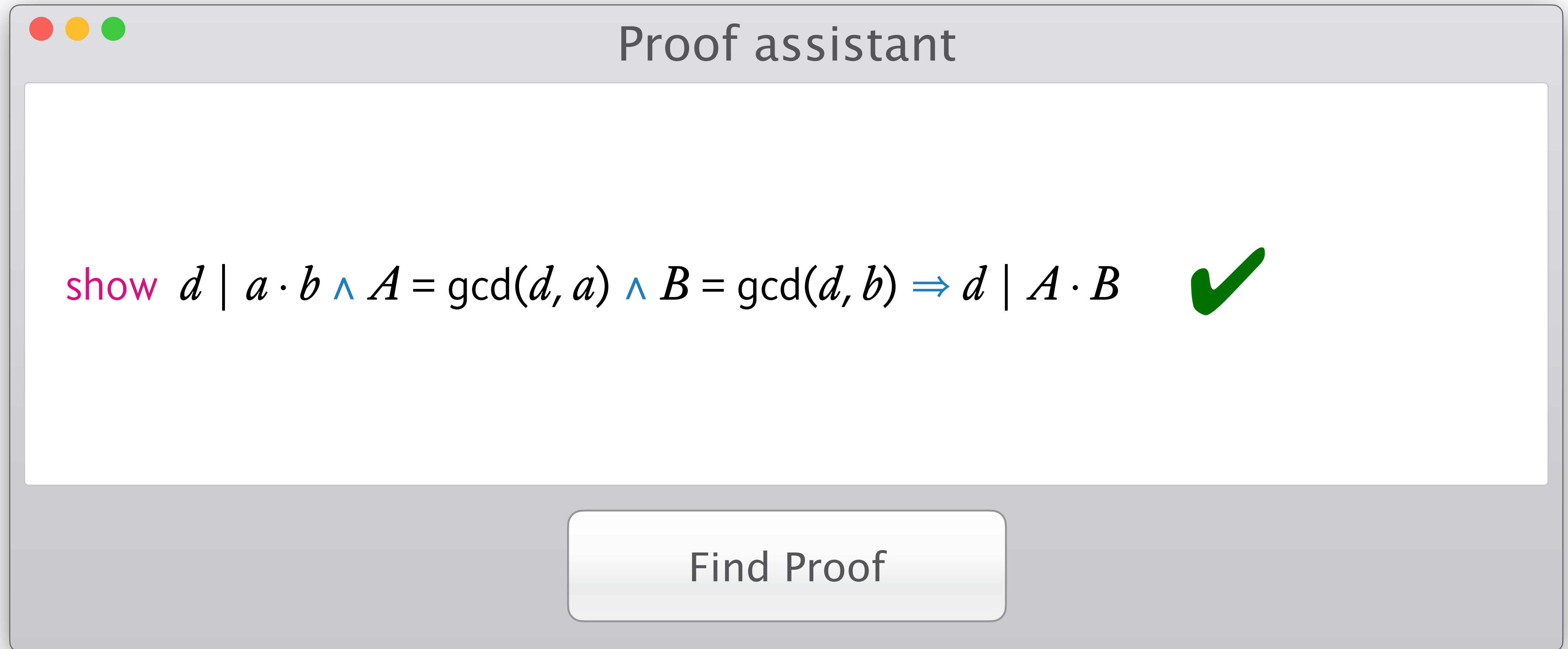
Provers and Solvers as Backends



Provers and Solvers as Backends



Provers and Solvers as Backends



Outline of These Lectures

- 1. Resolution**
2. Superposition
3. λ -Superposition
4. AVATAR

Proving by Saturation

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Proof goal: $F_1 \wedge \dots \wedge F_n \Rightarrow G$

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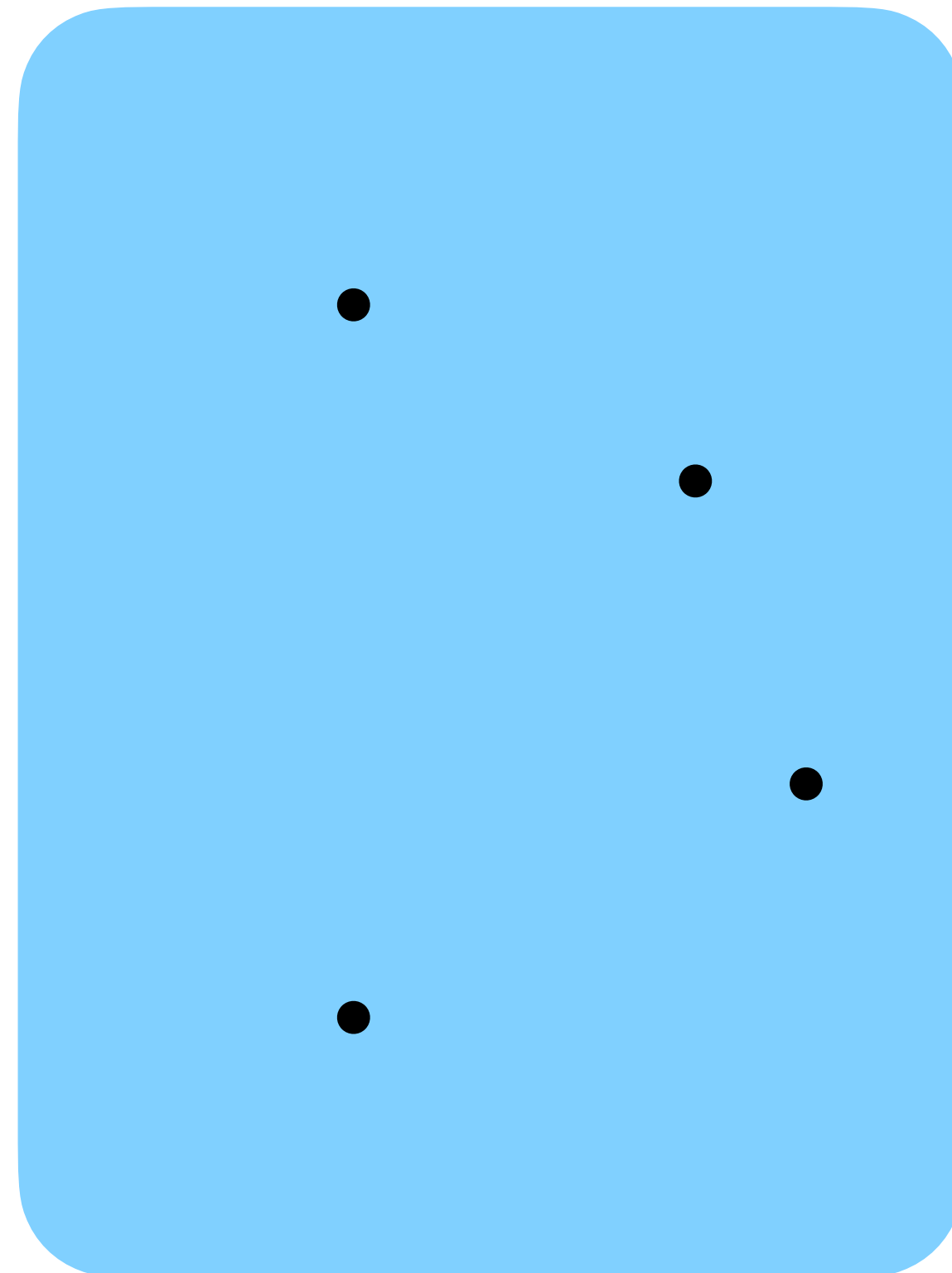
Proving by Saturation

Proof goal: $F_1 \wedge \dots \wedge F_n \Rightarrow G$

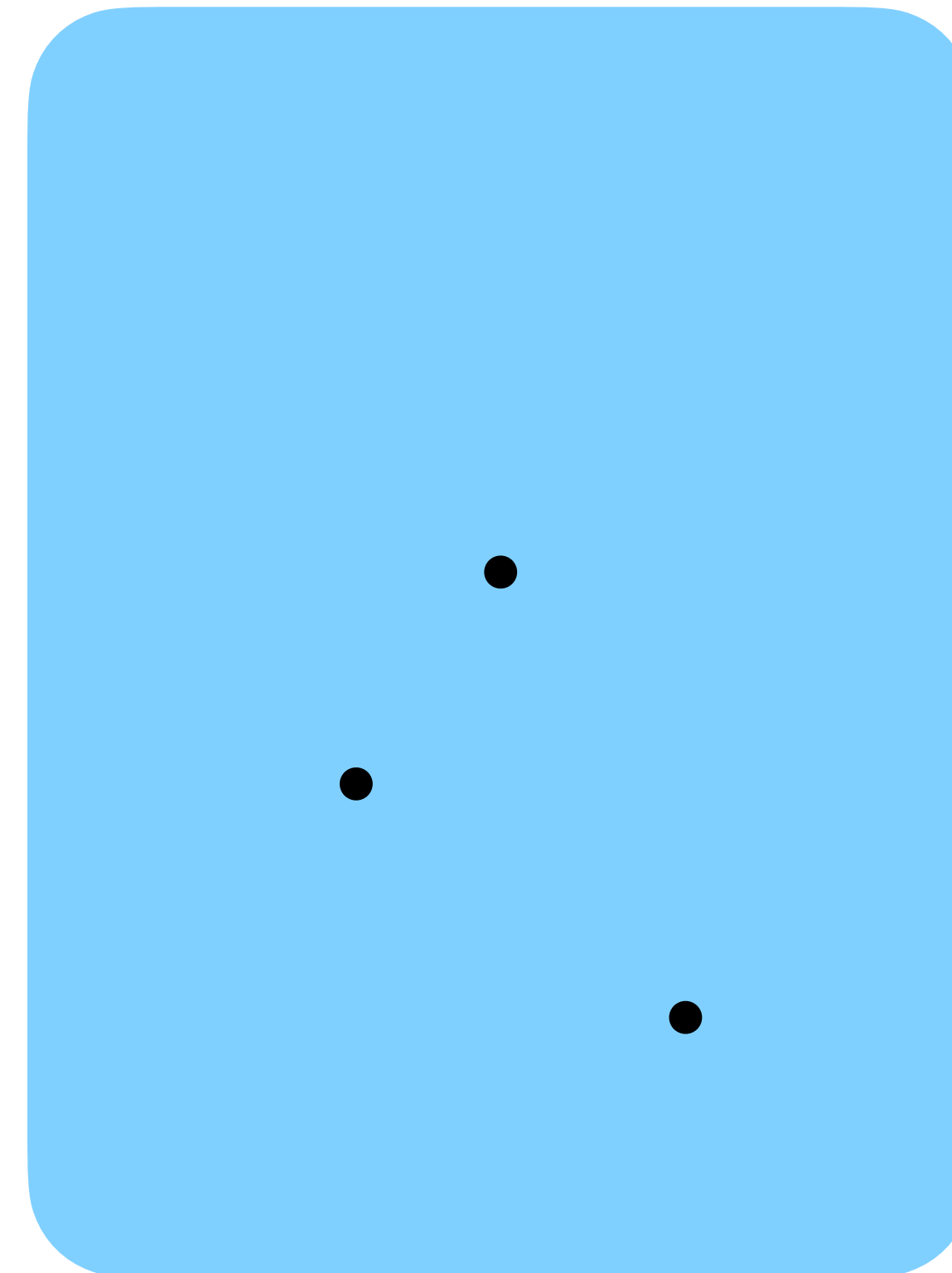
1. Put $F_1, \dots, F_n$ and the negation of G in **normal form**, yielding a formula set $\mathbb{F}$
2. Repeat:
 - 2.1. Apply an **inference rule** to $\mathbb{F}$ and add the conclusion to $\mathbb{F}$
 - 2.2. Possibly remove redundant formula
(e.g., $p \vee q$ in the presence of p)
 - 2.3. Stop when $\perp$ is in $\mathbb{F}$

Standard Saturation Loop

Passive

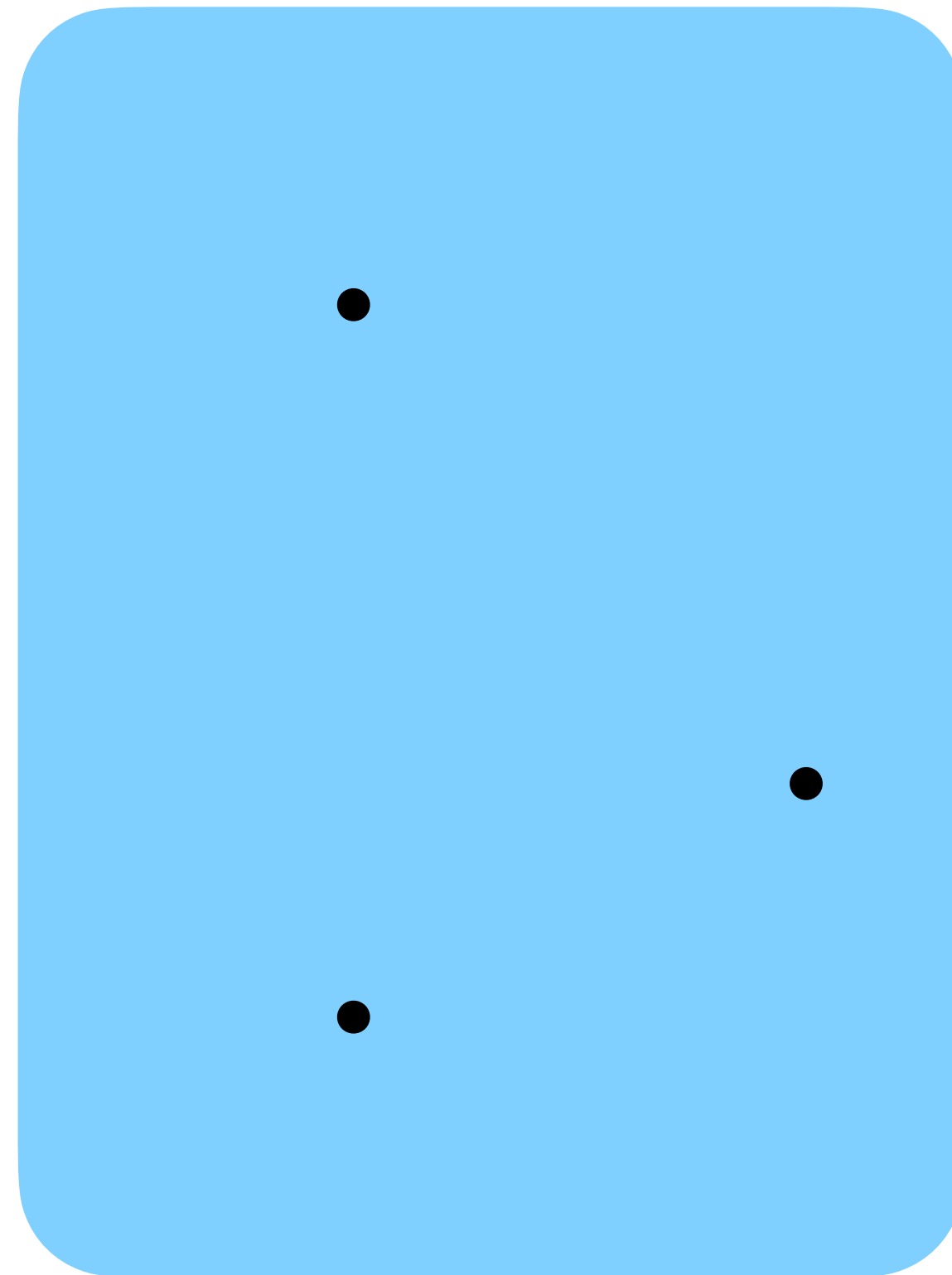


Active

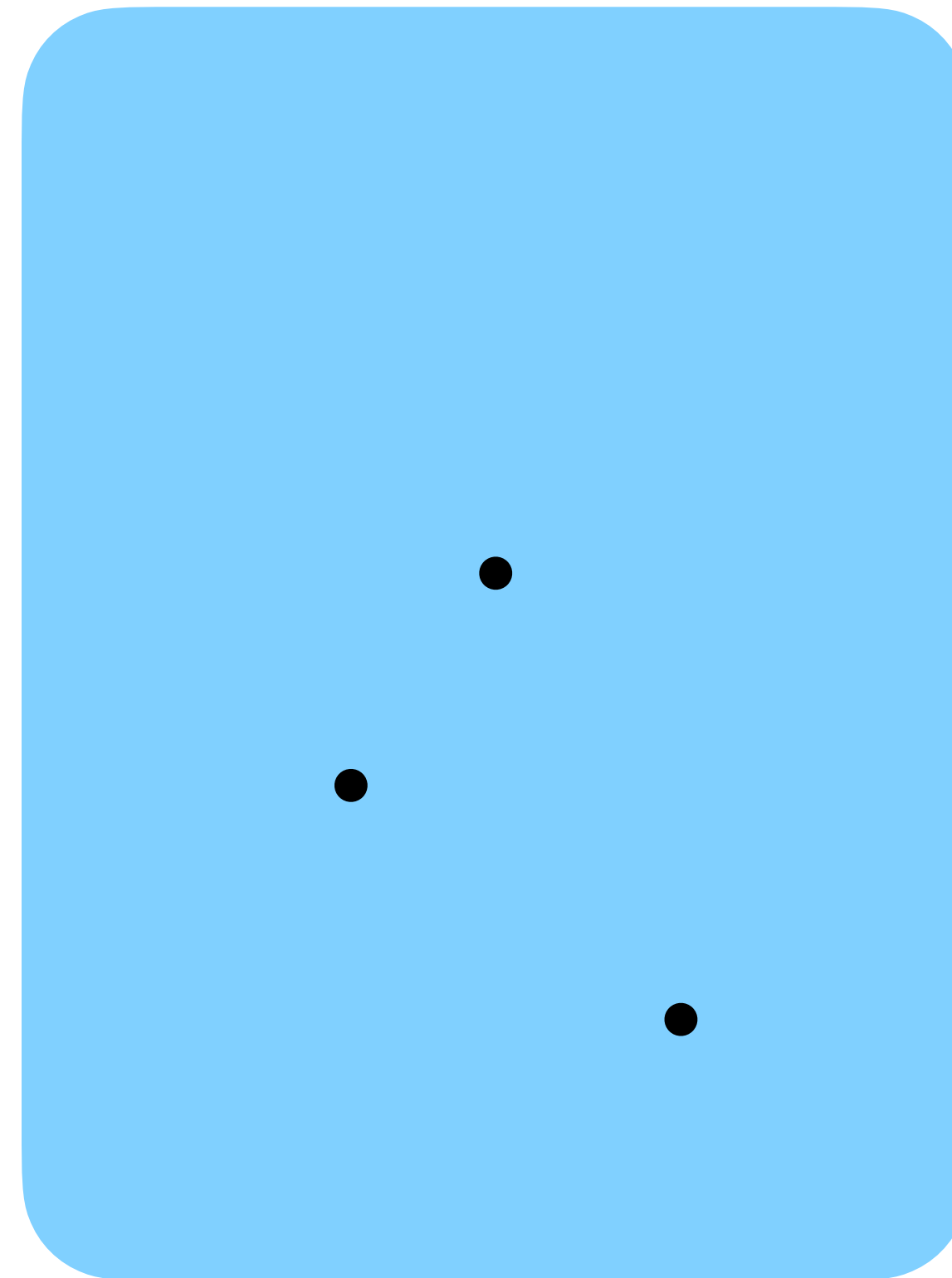


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Passive

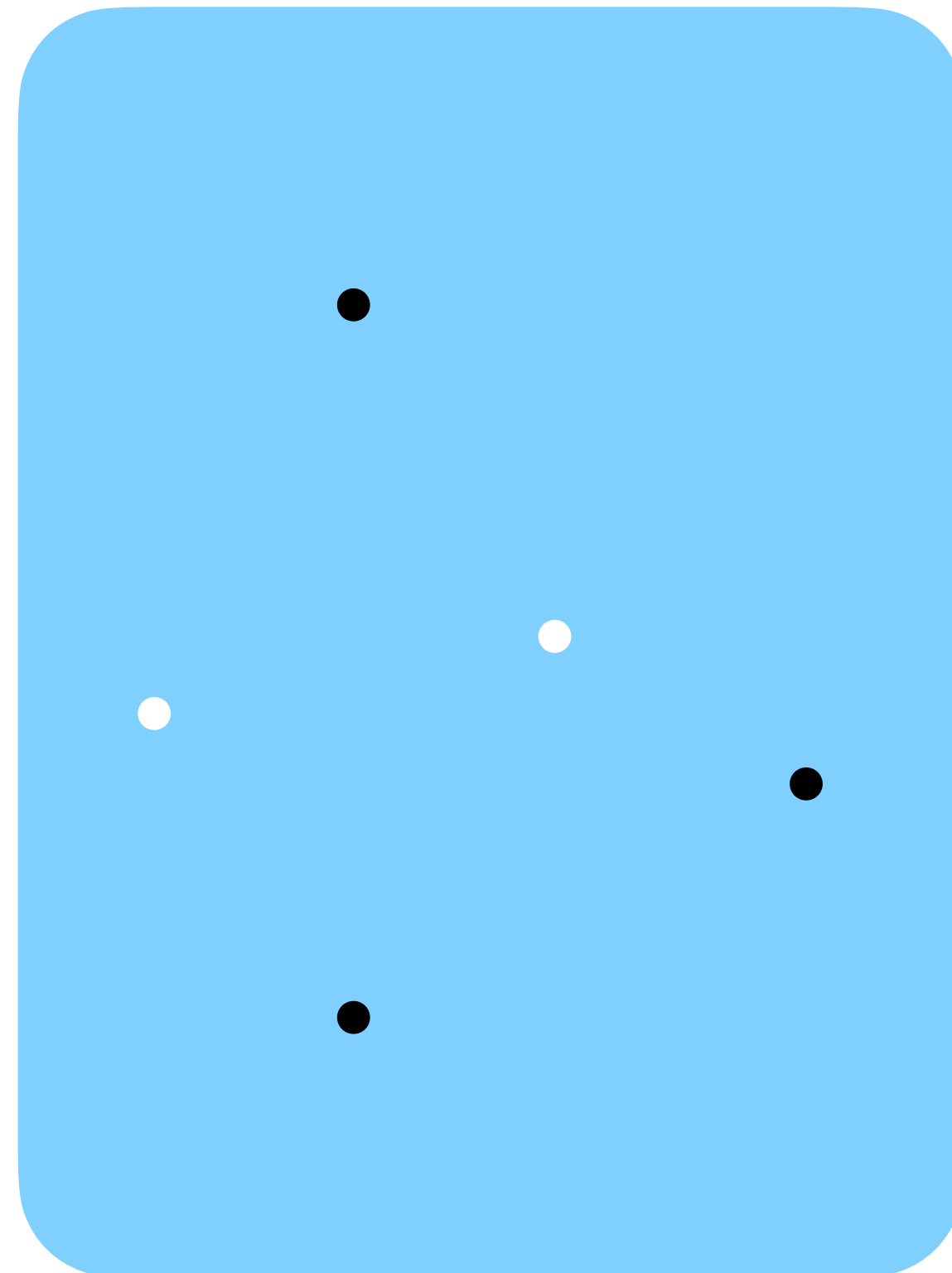


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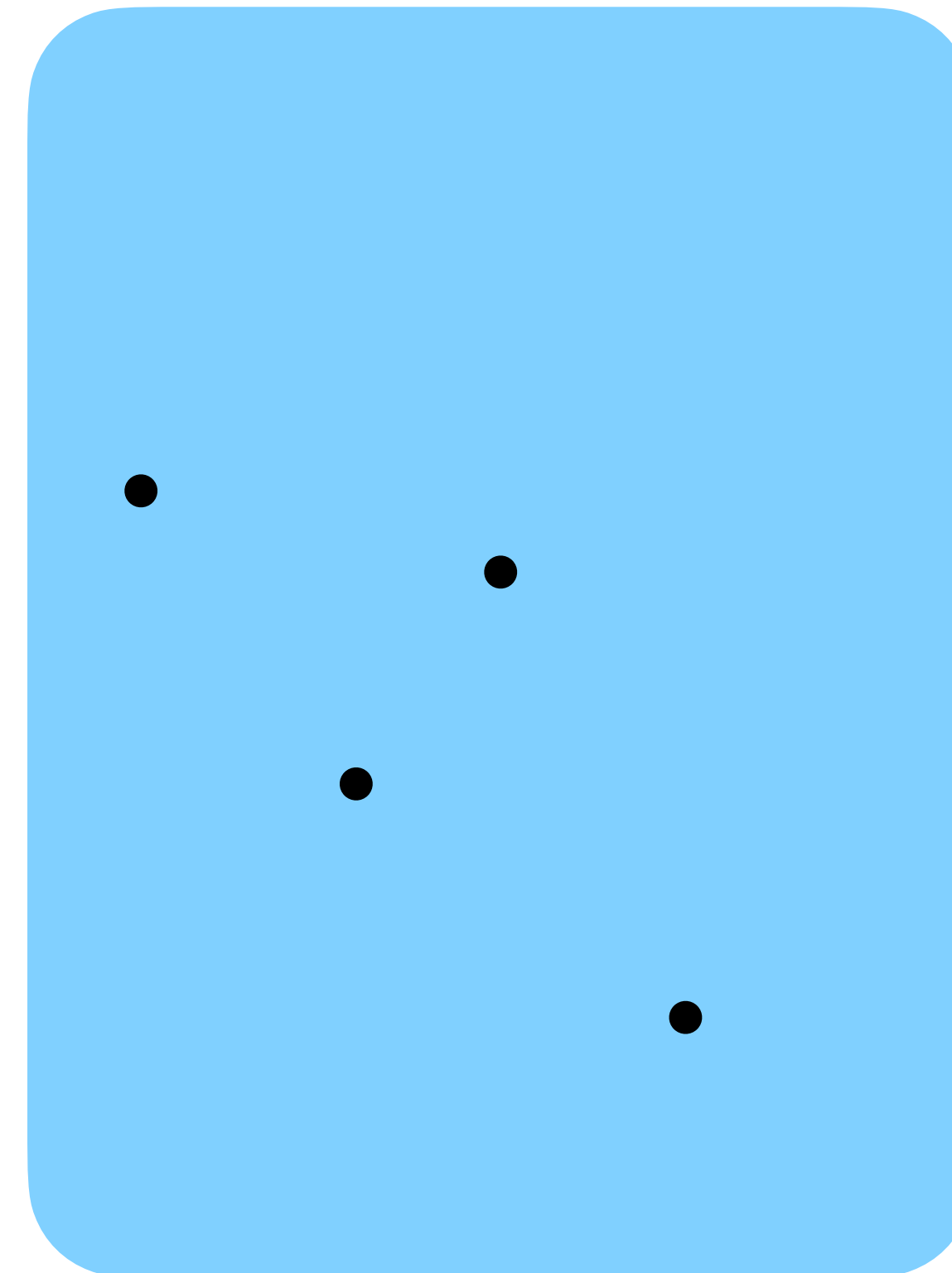


Standard Saturation Loop

Passive



Active



Syntax of First-Order (Predicate) Logic

Given a **signature**, consisting of

- function symbols (e.g., 1 , $\cdot$, gcd)
- predicate symbols (e.g., $=$, $|$, prime)

Syntax of First-Order (Predicate) Logic

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The **terms** are defined by these two rules:

- A variable x is a term
- If f is a function symbol and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term

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The **formulas** are defined by these rules:

- If p is a predicate symbol and $t_1, \dots, t_n$ are terms, then $p(t_1, \dots, t_n)$ is a formula
- If F and G are formulas and x is a variable, then $\perp$, $\top$, $\neg F$, $F \wedge G$, $F \vee G$, $F \Rightarrow G$, $\forall x. F$, and $\exists x. F$ are formulas

Example: A Puzzle in First-Order Logic

$(\forall x. \text{food}(x) \Rightarrow \text{likes}(\text{johanna}, x))$
 $\wedge (\forall x. (\exists y. \text{eats}(y, x) \wedge \neg \text{was_killed_by}(y, x)) \Rightarrow \text{food}(x))$
 $\wedge \text{eats}(\text{bill}, \text{peanuts}) \wedge \text{alive}(\text{bill})$
 $\wedge (\forall y. \text{alive}(y) \Rightarrow \forall x. \neg \text{was_killed_by}(y, x))$
 $\Rightarrow \text{likes}(\text{johanna}, \text{peanuts})$

Example: A Familiar Formula

$$d \mid a \cdot b \wedge A = \gcd(d, a) \wedge B = \gcd(d, b) \Rightarrow d \mid A \cdot B$$

Semantics of First-Order Logic

An **interpretation** $\mathcal{J}$ consists of

- a nonempty set D , called **domain**
- an operator $\mathcal{J}$ for the symbols:
 - $f^{\mathcal{J}} : D^n \rightarrow D$ for every n -ary function symbol f
 - $p^{\mathcal{J}} \subseteq D^n$ for every n -ary predicate symbol p

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The interpretation $\mathcal{J}$ is lifted to ground (variable-free) terms as follows:

$$\llbracket f(t_1, \dots, t_n) \rrbracket^{\mathcal{J}} = f^{\mathcal{J}}(\llbracket t_1 \rrbracket^{\mathcal{J}}, \dots, \llbracket t_n \rrbracket^{\mathcal{J}})$$

Semantics of First-Order Logic

The interpretation $\mathcal{J}$ is lifted to ground formulas as follows:

$\llbracket p(t_1, \dots, t_n) \rrbracket^{\mathcal{J}}$ iff $(\llbracket t_1 \rrbracket^{\mathcal{J}}, \dots, \llbracket t_n \rrbracket^{\mathcal{J}}) \in p^{\mathcal{J}}$

$\llbracket \perp \rrbracket^{\mathcal{J}}$ iff false

$\llbracket \top \rrbracket^{\mathcal{J}}$ iff true

$\llbracket \neg F \rrbracket^{\mathcal{J}}$ iff not $\llbracket F \rrbracket^{\mathcal{J}}$

$\llbracket F \wedge G \rrbracket^{\mathcal{J}}$ iff $\llbracket F \rrbracket^{\mathcal{J}}$ and $\llbracket G \rrbracket^{\mathcal{J}}$

etc.

Syntax of Clausal First-Order Logic

The **atoms** are defined by this rule:

- If p is a predicate symbol and $t_1, \dots, t_n$ are terms, then $p(t_1, \dots, t_n)$ is an atom

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- If A is an atom, then A and $\neg A$ are literals

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The **clauses** are defined by this rule:

- If $L_1, \dots, L_n$ are literals, then $L_1 \vee \dots \vee L_n$ (viewed as a multiset) is a clause

We write $\perp$ if $n = 0$

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We write $\perp$ if $n = 0$



- No $\top, \wedge, \Rightarrow, \forall, \exists$
- Variables are understood as "for all"

Clausification

Any formula can be turned into clauses using the following procedure:

1. Rewrite $F \Rightarrow G$ to $\neg F \vee G$
2. Push negations inwards using De Morgan's laws and eliminate double negations
3. Pull $\forall$ and $\exists$ outwards
4. Replace $\exists$ s with Skolem function symbols
5. Remove $\forall$ s
6. Distribute $\vee$ inwards over $\wedge$

Each conjunct is a clause

Example: Clausification

$(\forall x. \text{human}(x) \Rightarrow \text{mortal}(x))$
 $\wedge \text{human}(\text{anne})$
 $\Rightarrow \text{mortal}(\text{anne})$



$(\forall x. \text{human}(x) \Rightarrow \text{mortal}(x))$
 $\text{human}(\text{anne})$
 $\neg \text{mortal}(\text{anne})$

Example: Clausification

$(\forall x. \text{human}(x) \Rightarrow \text{mortal}(x))$
 $\wedge \text{human}(\text{anne})$
 $\Rightarrow \text{mortal}(\text{anne})$



$(\forall x. \neg \text{human}(x) \vee \text{mortal}(x))$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

Example: Clausification

$(\forall x. \text{human}(x) \Rightarrow \text{mortal}(x))$
 $\wedge \text{human}(\text{anne})$
 $\Rightarrow \text{mortal}(\text{anne})$



$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

Ground Resolution Rule

If the clauses

$$\begin{array}{l} C \vee A \\ \neg A \vee D \end{array}$$

are contained in $\mathbb{F}$,
then add the clause

$$C \vee D$$

to $\mathbb{F}$

Example: A Resolution Proof

F

$\neg \text{human}(\text{anne}) \vee \text{mortal}(\text{anne})$ $\text{human}(\text{anne})$ $\neg \text{mortal}(\text{anne})$

Example: A Resolution Proof

F

$\neg \text{human}(\text{anne}) \vee \text{mortal}(\text{anne})$ $\text{human}(\text{anne})$ $\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$

The diagram illustrates a resolution step. Two clauses are shown in a light gray rounded rectangle: $\neg \text{human}(\text{anne}) \vee \text{mortal}(\text{anne})$ and $\text{human}(\text{anne})$. Arrows point from the $\text{mortal}(\text{anne})$ literal in the first clause and the $\text{human}(\text{anne})$ literal in the second clause to a single $\text{mortal}(\text{anne})$ literal below, representing the resolvent.

Example: A Resolution Proof

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$\neg \text{human}(\text{anne}) \vee \text{mortal}(\text{anne})$ $\text{human}(\text{anne})$ $\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$

Example: A Resolution Proof

F

$\neg$ human(anne) $\vee$ mortal(anne)

human(anne)

$\neg$ mortal(anne)

mortal(anne)



Example: A Resolution Proof

F

$\neg \text{human}(\text{anne}) \vee \text{mortal}(\text{anne})$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$



Soundness of Ground Resolution

If the clauses

$$C \vee A$$

$$\neg A \vee D$$

are true,

then the clause

$$C \vee D$$

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			T	T	

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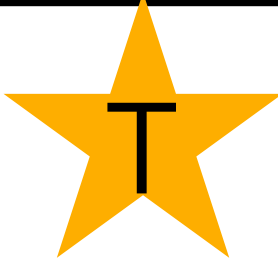
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F					

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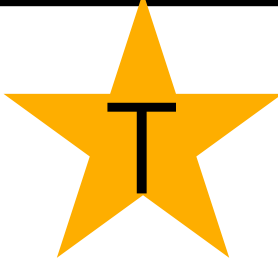
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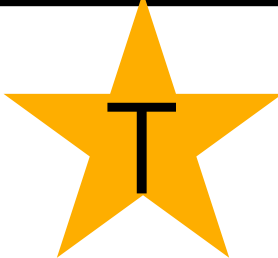
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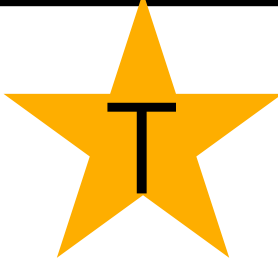
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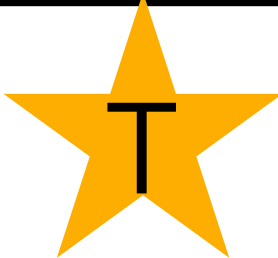
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Completeness of Ground Resolution

If the ground clause set $\mathcal{F}$ is initially unsatisfiable and inferences are performed fairly, then $\mathcal{F}$ will eventually contain $\perp$

Completeness of Ground Resolution

If the ground clause set $\mathcal{F}$ is initially unsatisfiable and inferences are performed fairly, then $\mathcal{F}$ will eventually contain $\perp$

Proof idea:

- By fairness, the limit is saturated (all inferences have been done)
- By "static completeness", the limit contains $\perp$

Static Completeness of Ground Resolution

If a clause set $\mathcal{F}$ is unsatisfiable and saturated,
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Static Completeness of Ground Resolution

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Contrapositive: If a clause set $\mathcal{F}$ is saturated
and does not contain $\perp$, then $\mathcal{F}$ has a model

Static Completeness of Ground Resolution

If a clause set $\mathcal{F}$ is unsatisfiable and saturated,
then $\mathcal{F}$ contains $\perp$

Contrapositive: If a clause set $\mathcal{F}$ is saturated
and does not contain $\perp$, then $\mathcal{F}$ has a model

Proof idea:

We build a **term model** incrementally
based on an order on the clauses

What Is a Term Model?

A term model is a model in which all terms are interpreted as themselves:

$$\llbracket f(t_1, \dots, t_n) \rrbracket^J = f(t_1, \dots, t_n)$$

What Is a Term Model?

A term model is a model in which all terms are interpreted as themselves:

$$\llbracket f(t_1, \dots, t_n) \rrbracket^J = f(t_1, \dots, t_n)$$

Term models can be represented as the set of ground atoms that are true

E.g. $\{p(f(a))\}$ makes $p(f(a))$ true and everything else false

Static Completeness of Ground Resolution

Contrapositive: If a clause set $\mathcal{F}$ is saturated and does not contain $\perp$, then $\mathcal{F}$ has a model

$\mathcal{F}$

$q \vee r$

$\neg q \vee \neg t \vee s$

$\neg q \vee t \vee s$

$\neg q \vee r$

$p \vee q$

Static Completeness of Ground Resolution

Contrapositive: If a clause set $\mathcal{F}$ is saturated and does not contain $\perp$, then $\mathcal{F}$ has a model

Clause	Model	Δ	Remark
$p \vee q$ $q \vee r$ $\neg q \vee r$ $\neg q \vee t \vee s$ $\neg q \vee \neg t \vee s$			

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Clause	Model	Δ	Remark
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Static Completeness of Ground Resolution

Contrapositive: If a clause set $\mathcal{F}$ is saturated and does not contain $\perp$, then $\mathcal{F}$ has a model

Clause	Model	Δ	Remark
$p \vee \boxed{q}$	$\{\}$	$\{q\}$	q is maximal already true
$q \vee \boxed{r}$	$\{q\}$	$\{\}$	
$\neg q \vee \boxed{r}$			
$\neg q \vee \boxed{t} \vee s$			
$\neg q \vee \boxed{\neg t} \vee s$			

Static Completeness of Ground Resolution

Contrapositive: If a clause set $\mathcal{F}$ is saturated and does not contain $\perp$, then $\mathcal{F}$ has a model

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$\neg q \vee \boxed{t} \vee s$			
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$q \vee \boxed{r}$	$\{q\}$	$\{\}$	already true
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$\neg q \vee \boxed{t} \vee s$	$\{q, r\}$	$\{t\}$	t is maximal
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$\neg q \vee \boxed{\neg t} \vee s$	$\{q, r, t\}$	$\{\}$	???

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$\neg q \vee \neg q \vee s \vee s$

Static Completeness of Ground Resolution

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Clause	Model	Δ	Remark
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$\neg q \vee \boxed{\neg t} \vee s$	$\{q, r, t\}$	$\{\}$	???

$\neg q \vee \neg q \vee s \vee s$ ⚡

A Bad Scenario for Ground Resolution

$\mathcal{F}$

$a \vee b \vee c \vee d \vee e \quad \neg a \quad \neg b \quad \neg c \quad \neg d \quad \neg e$

A Bad Scenario for Ground Resolution

$\mathcal{F}$

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$b \vee c \vee d \vee e$

$a \vee c \vee d \vee e$

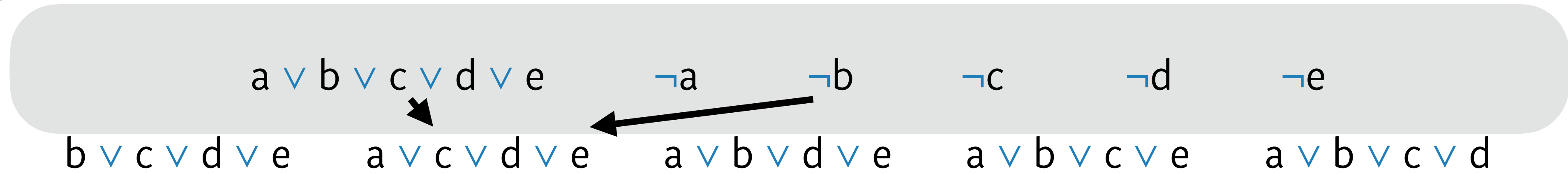
$a \vee b \vee d \vee e$

$a \vee b \vee c \vee e$

$a \vee b \vee c \vee d$

A Bad Scenario for Ground Resolution

F



A Bad Scenario for Ground Resolution

$\mathbb{F}$

$a \vee b \vee c \vee d \vee e$	$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$	

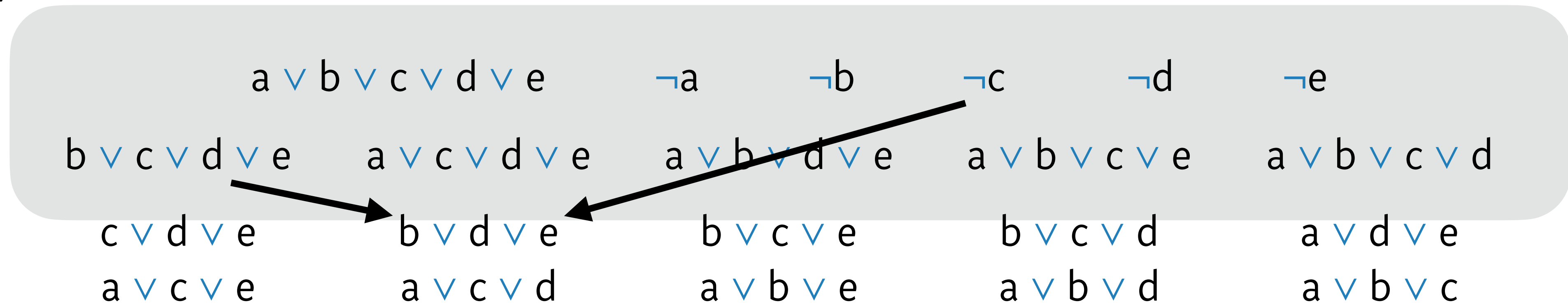
A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$	$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$	
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$	
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$	

A Bad Scenario for Ground Resolution

F



A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$		$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$		
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$		
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$		

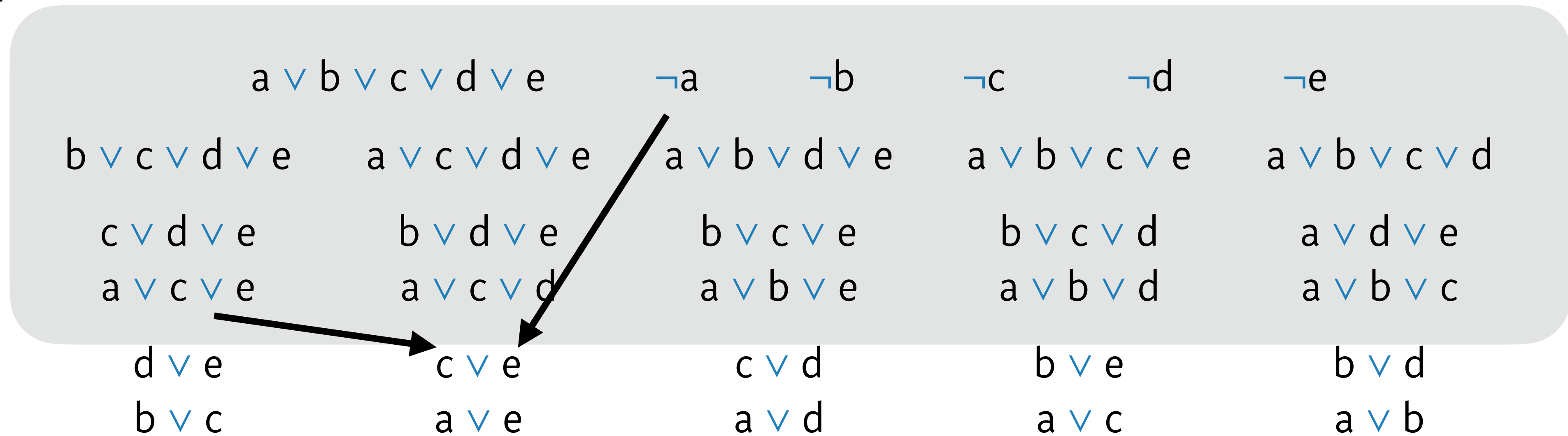
A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$		$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$		
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$		
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$		
$d \vee e$	$c \vee e$	$c \vee d$	$b \vee e$	$b \vee d$		
$b \vee c$	$a \vee e$	$a \vee d$	$a \vee c$	$a \vee b$		

A Bad Scenario for Ground Resolution

F



A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$		$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$		
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$		
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$		
$d \vee e$	$c \vee e$	$c \vee d$	$b \vee e$	$b \vee d$		
$b \vee c$	$a \vee e$	$a \vee d$	$a \vee c$	$a \vee b$		

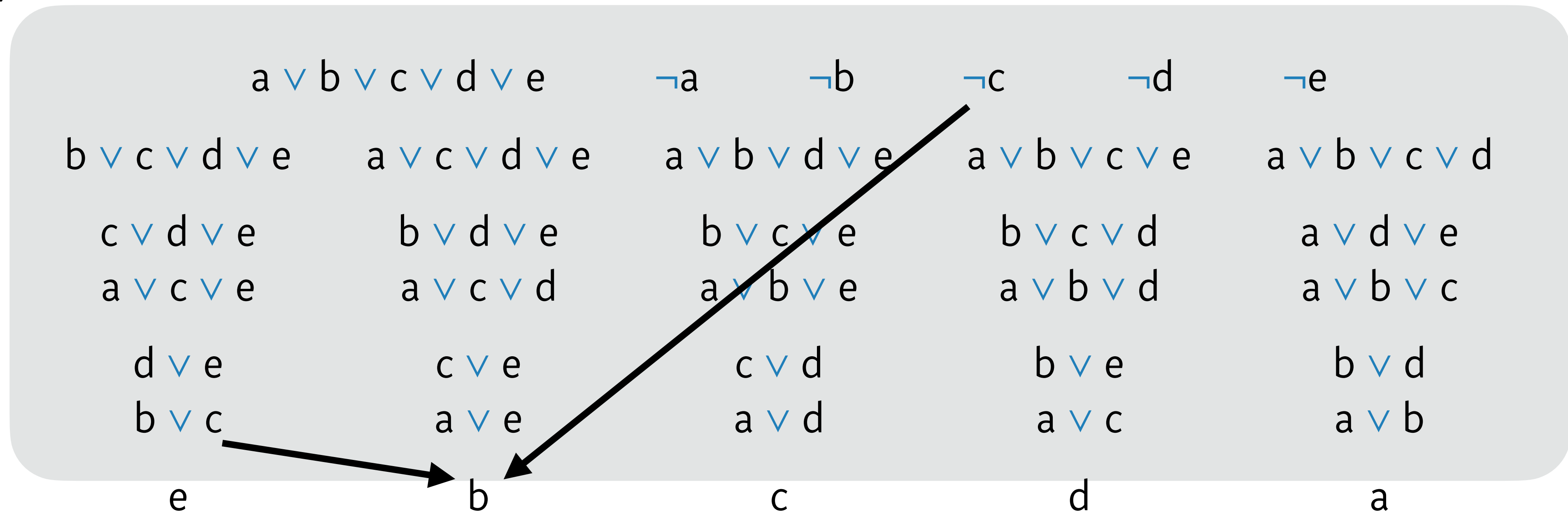
A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$					$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$					
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$					
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$					
$d \vee e$	$c \vee e$	$c \vee d$	$b \vee e$	$b \vee d$					
$b \vee c$	$a \vee e$	$a \vee d$	$a \vee c$	$a \vee b$					
e	b	c	d	a					

A Bad Scenario for Ground Resolution

F



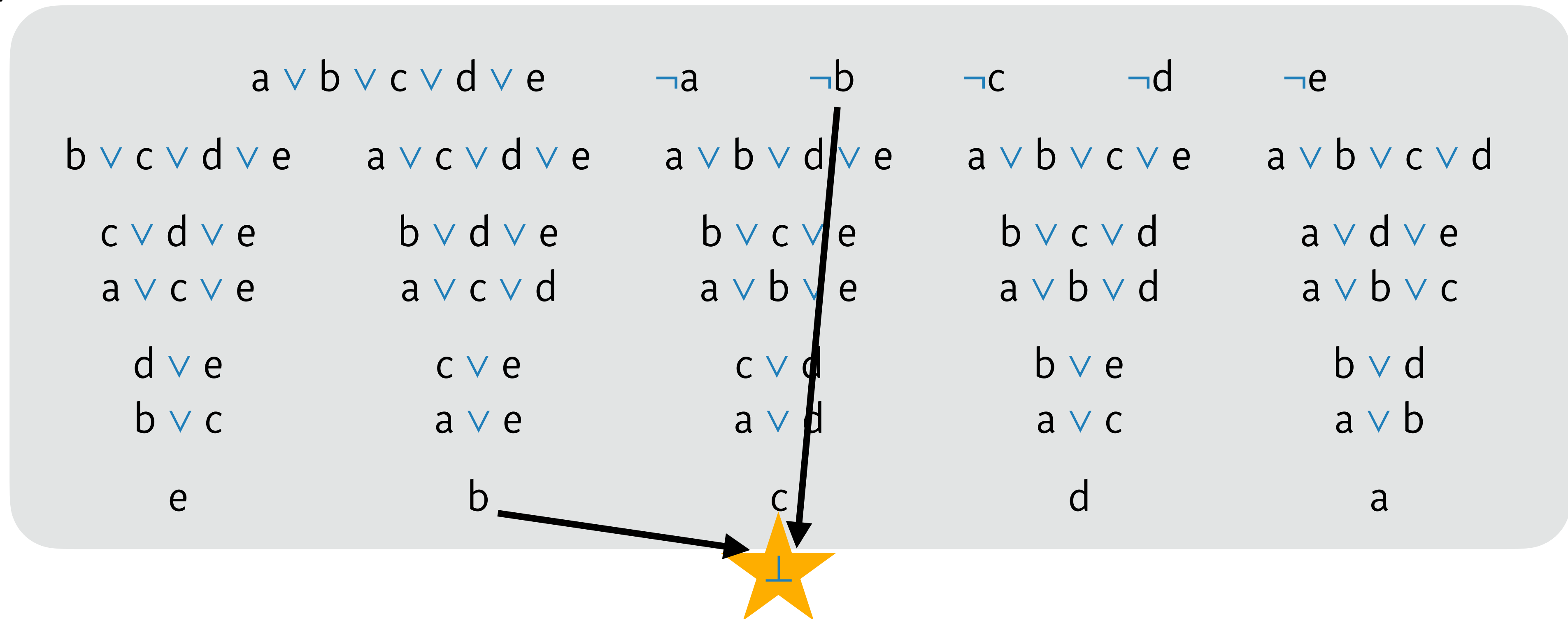
A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$		$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$		
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$		
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$		
$d \vee e$	$c \vee e$	$c \vee d$	$b \vee e$	$b \vee d$		
$b \vee c$	$a \vee e$	$a \vee d$	$a \vee c$	$a \vee b$		
e	b	c	d	a		

A Bad Scenario for Ground Resolution

F



A Bad Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$		$\neg a$	$\neg b$	$\neg c$	$\neg d$	$\neg e$
$b \vee c \vee d \vee e$	$a \vee c \vee d \vee e$	$a \vee b \vee d \vee e$	$a \vee b \vee c \vee e$	$a \vee b \vee c \vee d$		
$c \vee d \vee e$	$b \vee d \vee e$	$b \vee c \vee e$	$b \vee c \vee d$	$a \vee d \vee e$		
$a \vee c \vee e$	$a \vee c \vee d$	$a \vee b \vee e$	$a \vee b \vee d$	$a \vee b \vee c$		
$d \vee e$	$c \vee e$	$c \vee d$	$b \vee e$	$b \vee d$		
$b \vee c$	$a \vee e$	$a \vee d$	$a \vee c$	$a \vee b$		
e	b	c	d	a		
						

A Good Scenario for Ground Resolution

$\mathcal{F}$

$a \vee b \vee c \vee d \vee e \quad \neg a \quad \neg b \quad \neg c \quad \neg d \quad \neg e$

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$ $\neg a$ $\neg b$ $\neg c$ $\neg d$ $\neg e$

$a \vee b \vee c \vee d$

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

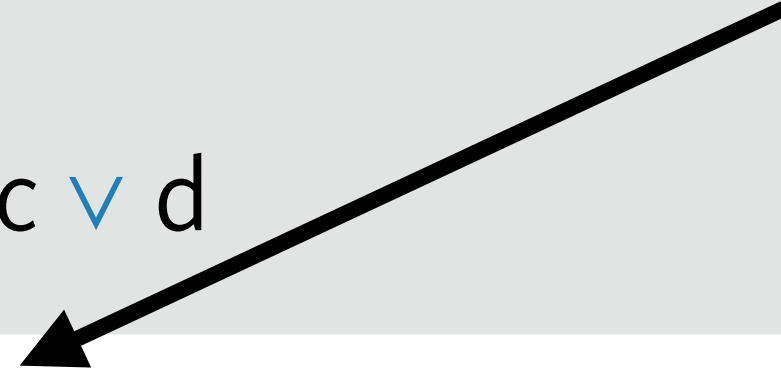
$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$



A Good Scenario for Ground Resolution

$\mathcal{F}$

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

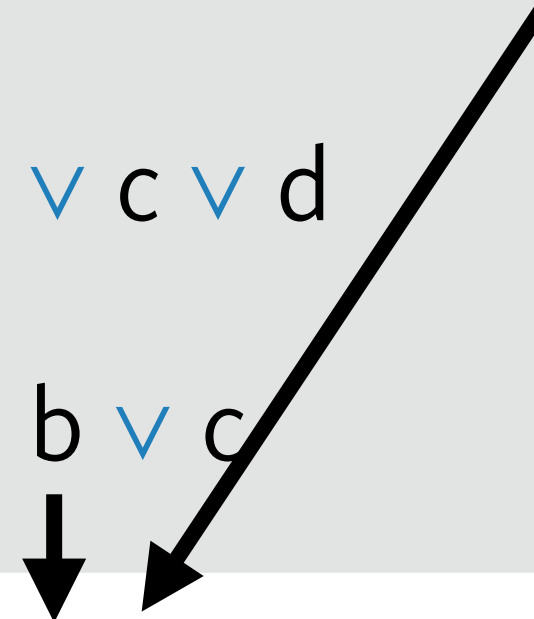
$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$



A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$

a

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$

a

A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$

a



A Good Scenario for Ground Resolution

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

$a \vee b \vee c$

$a \vee b$

a



Ground Ordered Resolution Rule

If the clauses

$$\begin{array}{c} C \vee \boxed{A} \\ \boxed{\neg A} \vee D \end{array}$$

are contained in $\mathbb{F}$
and A is **maximal** in $C \vee A$
and $\neg A$ is **maximal** in $\neg A \vee D$,
then add the clause

$$C \vee D$$

to $\mathbb{F}$

Soundness of Ordered Ground Resolution

If the clauses $C \vee A$ and $\neg A \vee D$ are true,
then the clause $C \vee D$ is true,
regardless of order restrictions

Soundness of Ordered Ground Resolution

If the clauses $C \vee A$ and $\neg A \vee D$ are true,
then the clause $C \vee D$ is true,
regardless of order restrictions

Proof: Special case of soundness of
(unordered) ground resolution

Completeness of Ordered Ground Resolution

If the clause set $\mathcal{F}$ is initially unsatisfiable
and inferences are performed fairly,
then $\mathcal{F}$ will eventually contain $\perp$

Completeness of Ordered Ground Resolution

If the clause set $\mathcal{F}$ is initially unsatisfiable and inferences are performed fairly, then $\mathcal{F}$ will eventually contain $\perp$

Proof idea:

- The proof for (unordered) ground resolution goes through

Static Completeness of Ordered Ground Resolution

Clause	Model	Δ	Remark
$\neg q \vee t \vee s$ $\neg q \vee \neg t \vee s$	$\{q, r\}$ $\{q, r, t\}$	$\{t\}$ $\{\}$	t is maximal ???

$\neg q \vee \neg q \vee s \vee s$



Ground Redundancy Criterion

A ground clause C is redundant w.r.t. ground $\mathcal{F}$ if $C_1, \dots, C_n \in \mathcal{F}$ are all smaller than C and they together entail ("imply") C

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Examples:

- a makes $a \vee b$ redundant (subsumption)
- $\boxed{b} \vee a$ and $\neg a \vee \boxed{c}$ make $b \vee c$ redundant

Ground Redundancy Criterion

A ground clause C is redundant w.r.t. ground $\mathcal{F}$ if $C_1, \dots, C_n \in \mathcal{F}$ are all smaller than C and they together entail ("imply") C

Examples:

- a makes $a \vee b$ redundant (subsumption)
- $\boxed{b} \vee a$ and $\neg a \vee \boxed{c}$ make $b \vee c$ redundant

The completeness proof still goes through (why?)

Static Completeness of Ordered Ground Resolution

Clause	Model	Δ	Remark
$\neg q \vee t \vee s$ $\neg q \vee \neg t \vee s$	$\{q, r\}$ $\{q, r, t\}$	$\{t\}$ $\{\}$	t is maximal ???

$\neg q \vee \neg q \vee s \vee s$



Example: Redundancy by Subsumption

F

$a \vee b \vee c \vee d \vee e$ $\neg a$ $\neg b$ $\neg c$ $\neg d$ $\neg e$

Example: Redundancy by Subsumption

F

$a \vee b \vee c \vee d \vee e$

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$

Example: Redundancy by Subsumption

F

~~$a \vee b \vee c \vee d \vee e$~~

$\neg a$

$\neg b$

$\neg c$

$\neg d$

$\neg e$

$a \vee b \vee c \vee d$



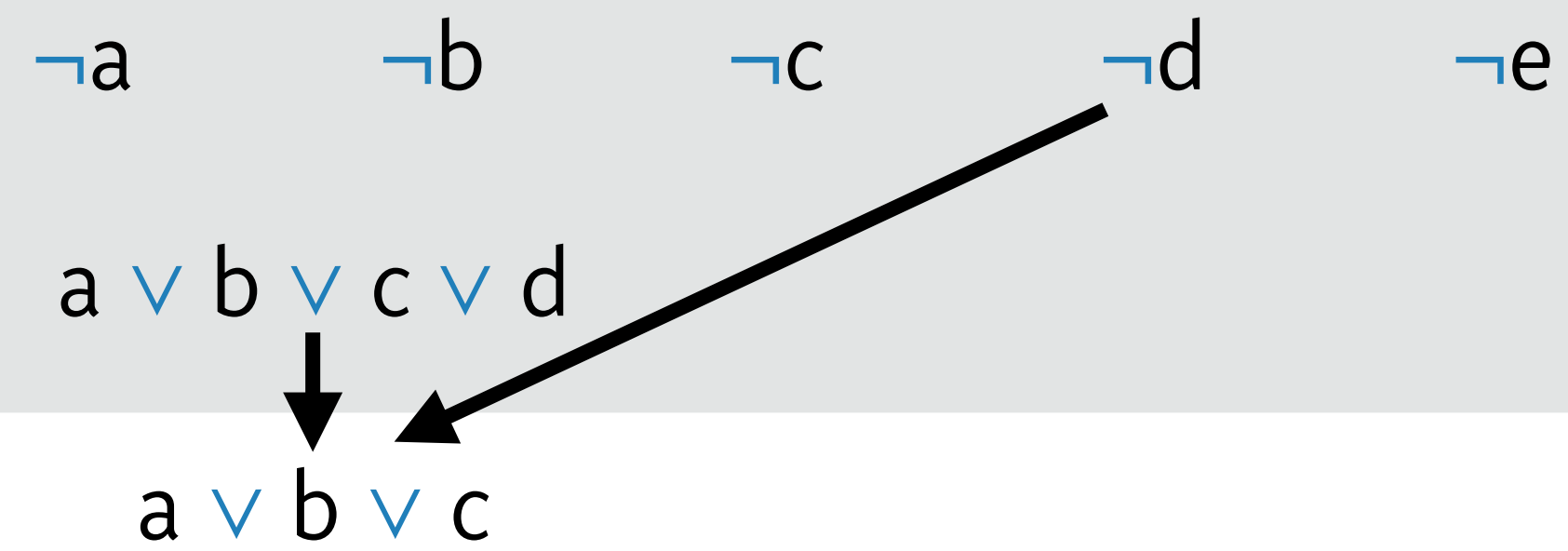
Example: Redundancy by Subsumption

F

$\neg a \quad \neg b \quad \neg c \quad \neg d \quad \neg e$
 $a \vee b \vee c \vee d$

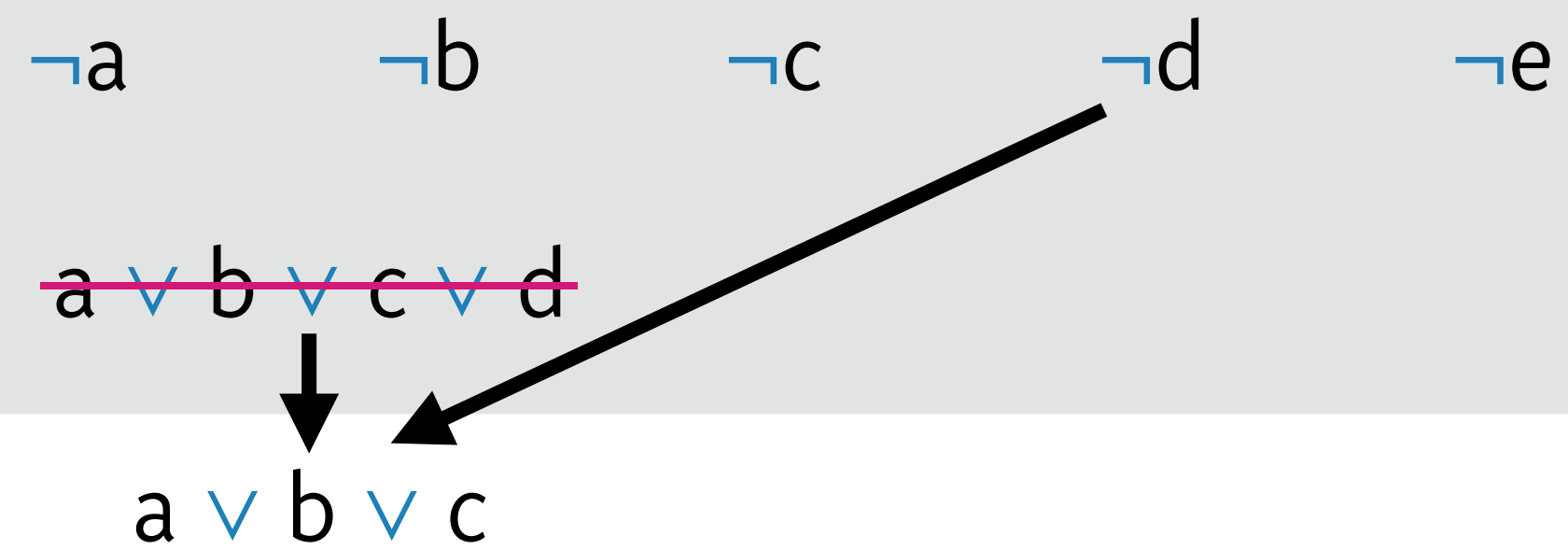
Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

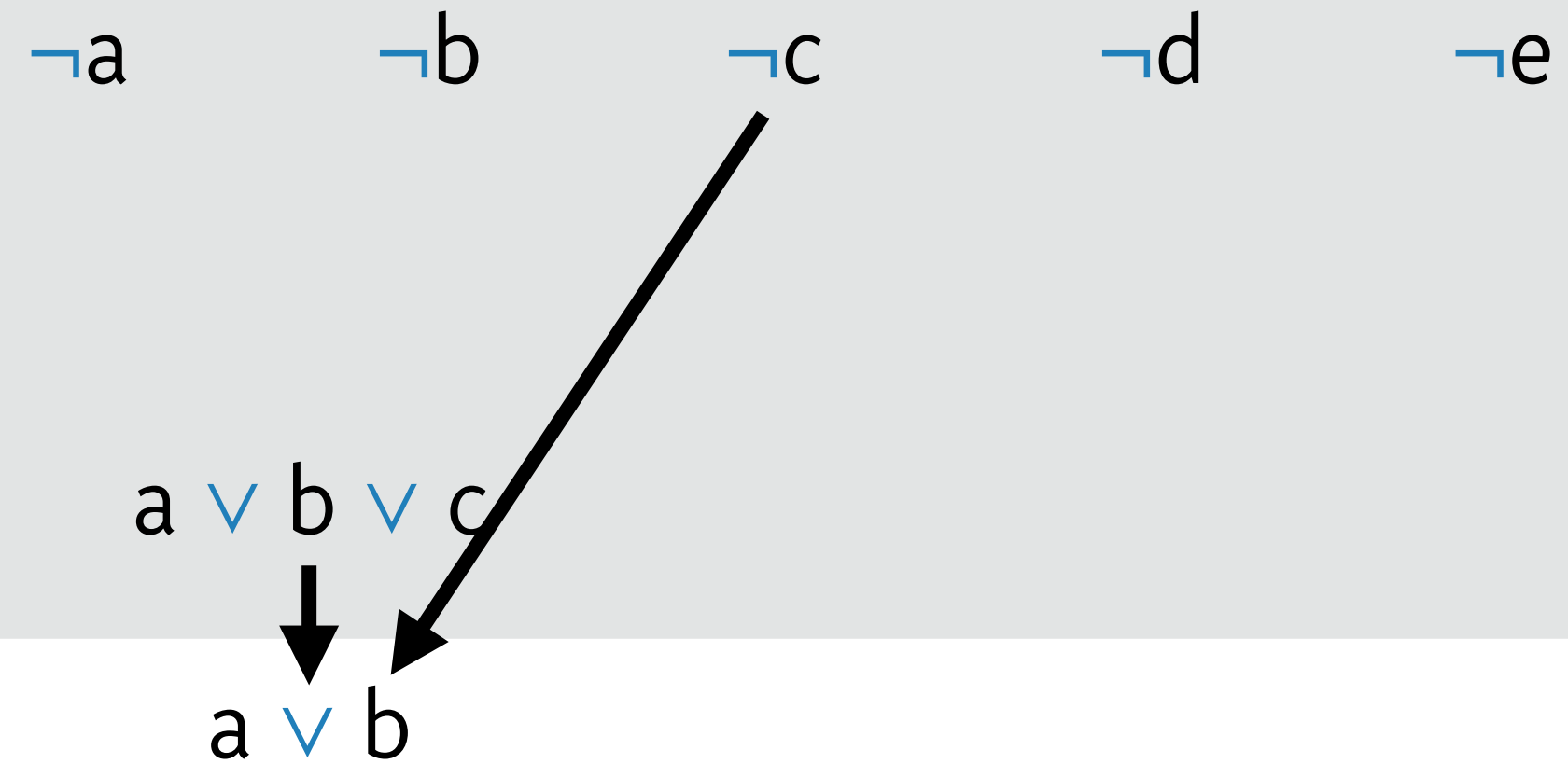
F

$\neg a$ $\neg b$ $\neg c$ $\neg d$ $\neg e$

$a \vee b \vee c$

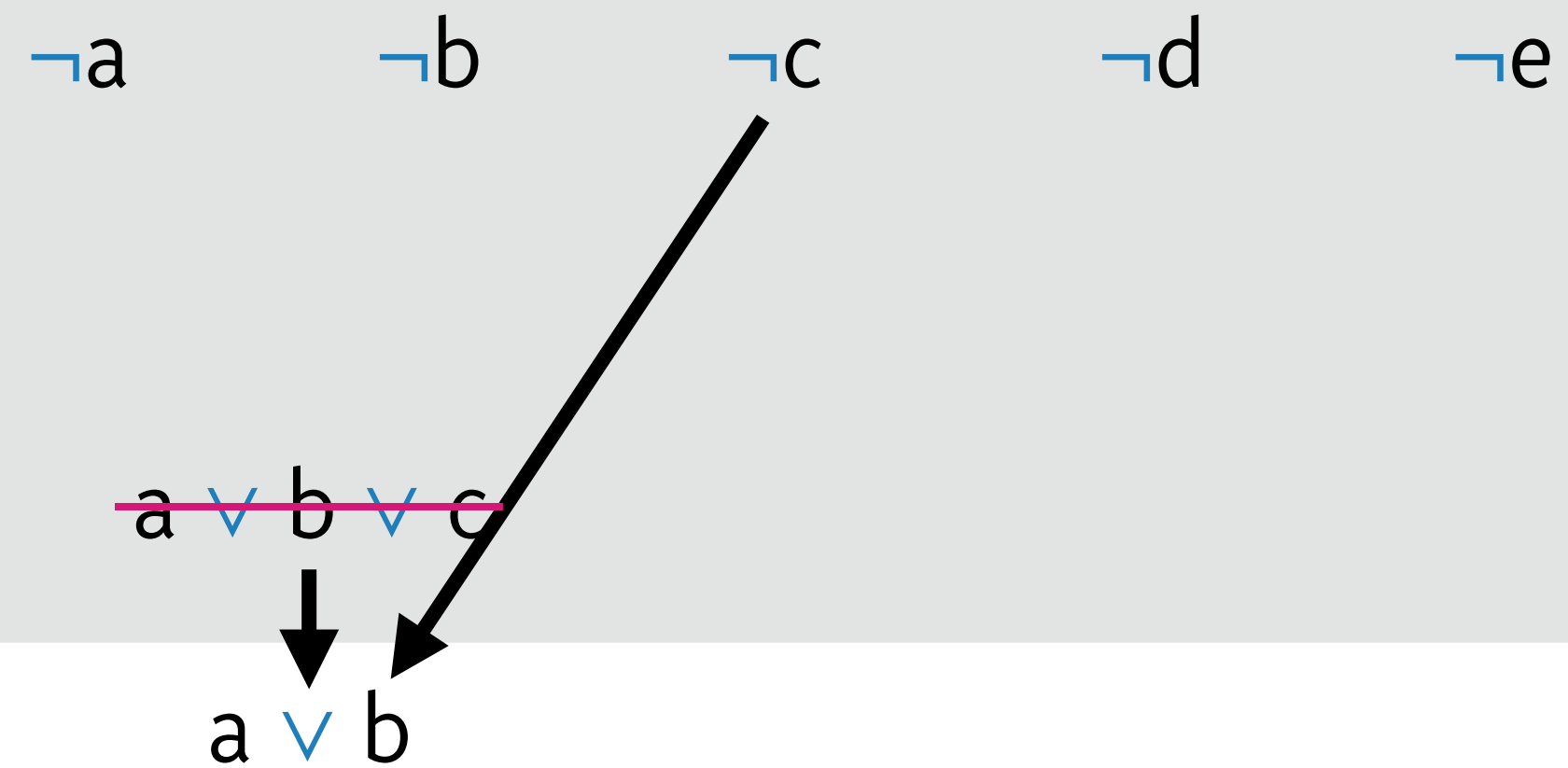
Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

F

$\neg a$ $\neg b$ $\neg c$ $\neg d$ $\neg e$

$a \vee b$

Example: Redundancy by Subsumption

F

$\neg a$

$\neg b$

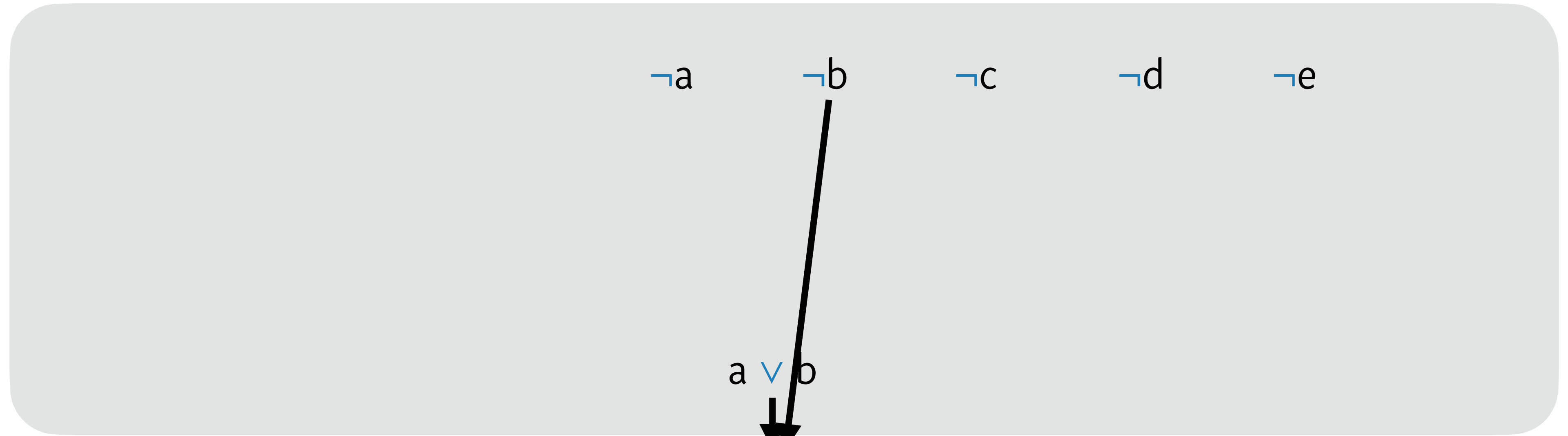
$\neg c$

$\neg d$

$\neg e$

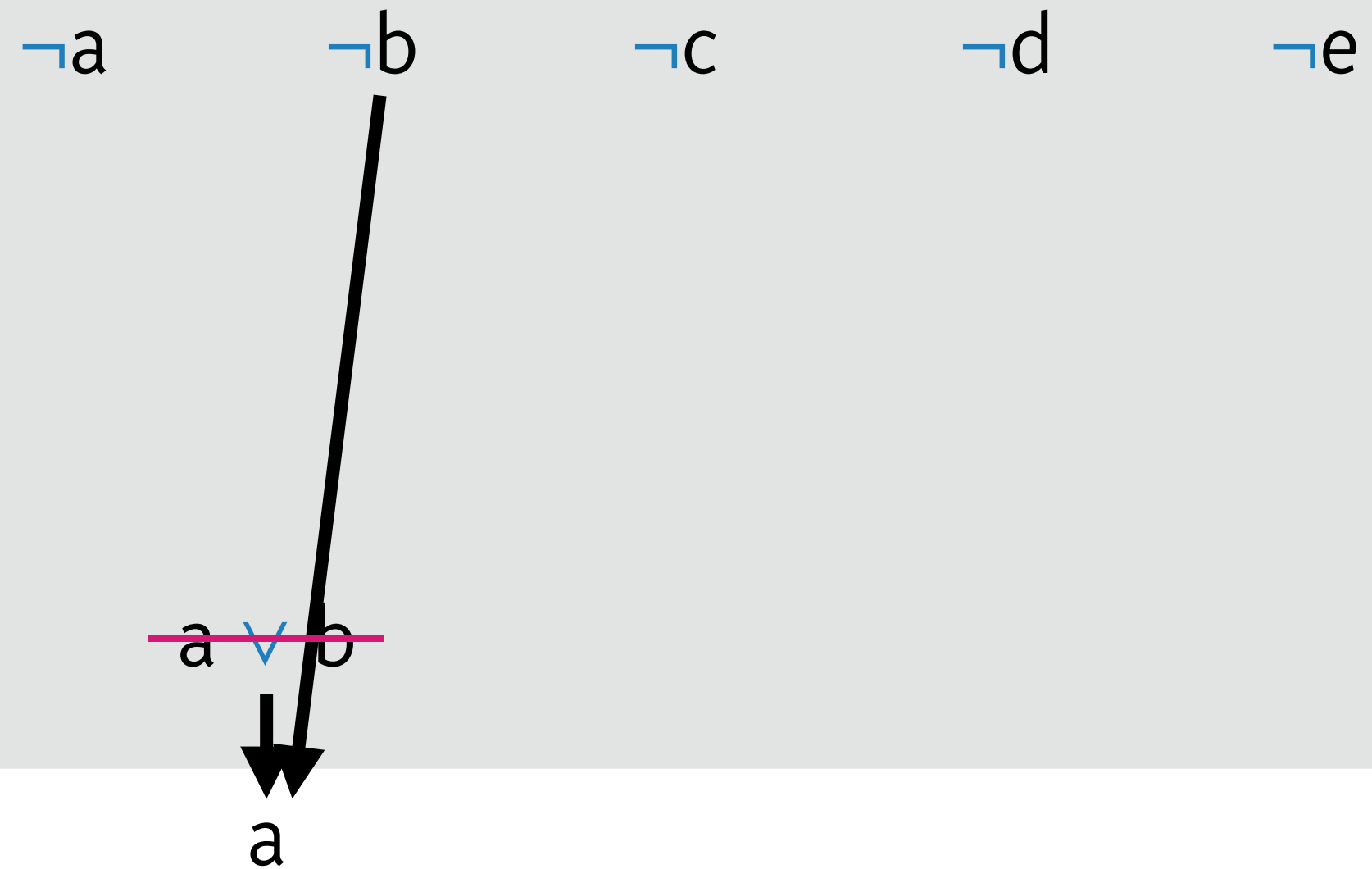
$a \vee b$

a



Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

F

$\neg a$

$\neg b$

$\neg c$

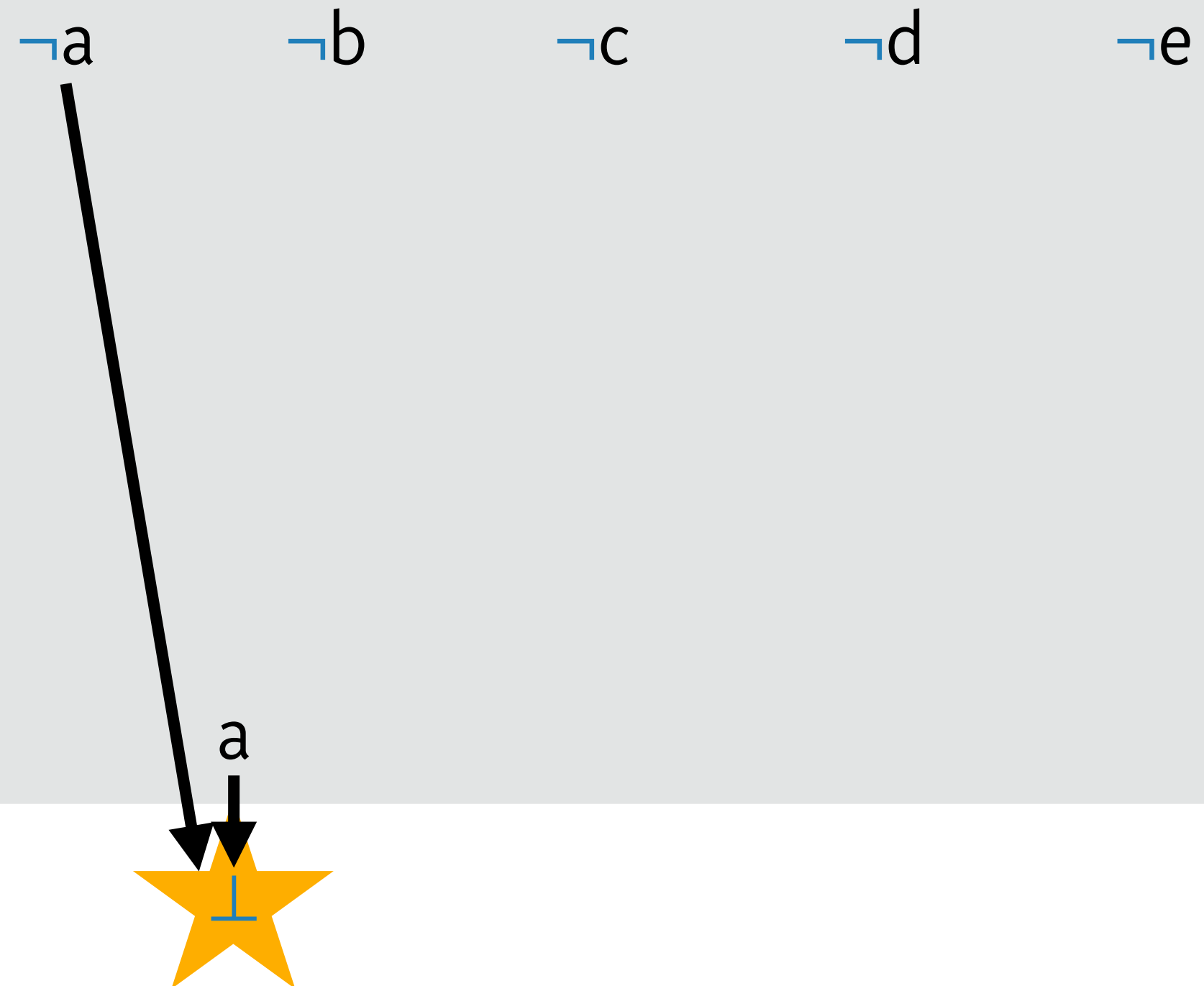
$\neg d$

$\neg e$

a

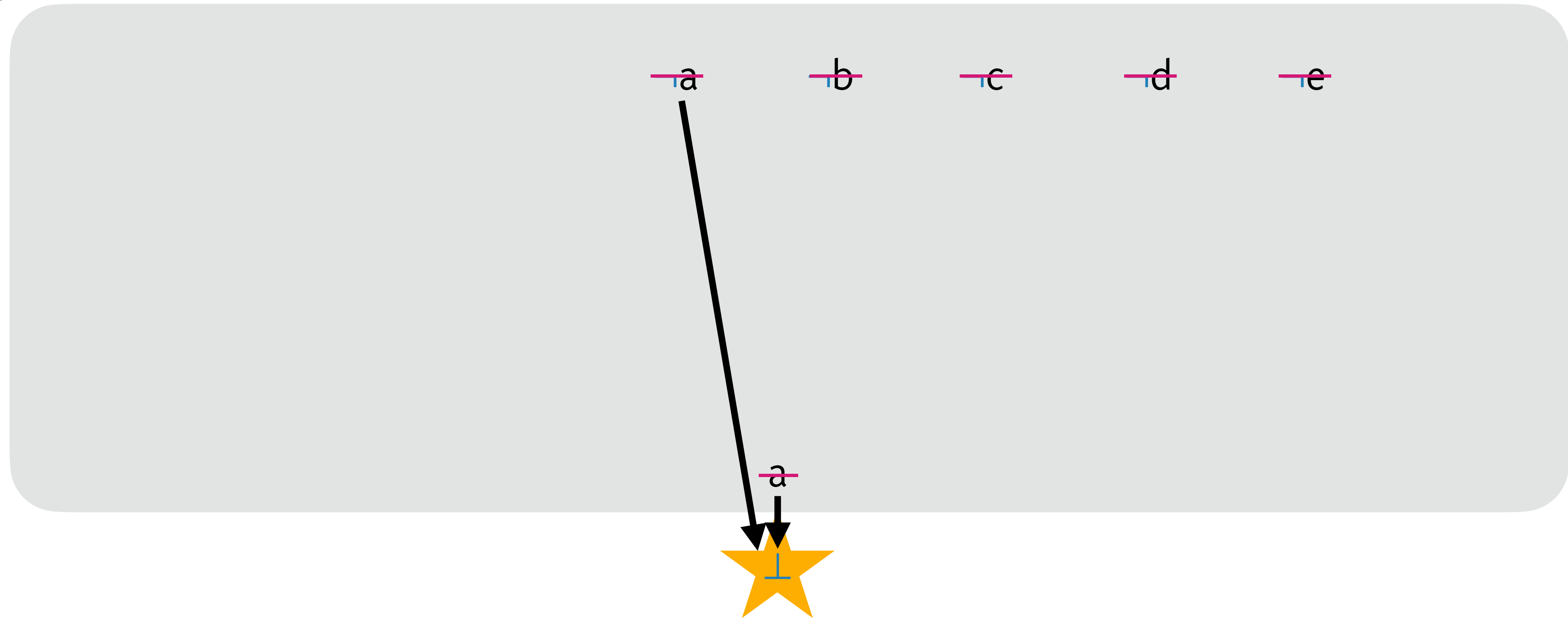
Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

F



Example: Redundancy by Subsumption

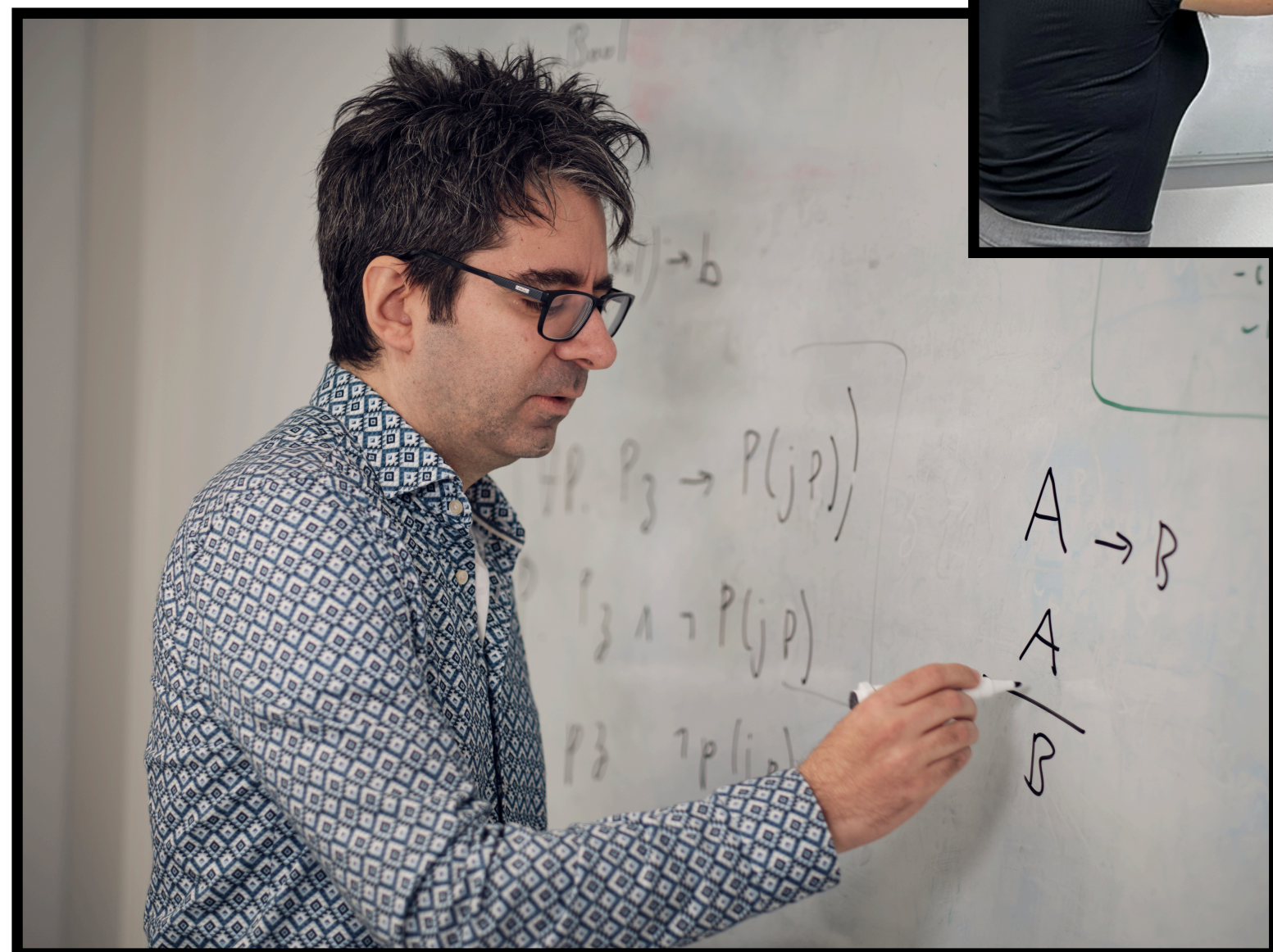
F



Who I Am

1978	Birth
2000	BSc from U Sherbrooke
	Mysterious gap
2008	MSc from U Oslo
2012	PhD from TU Munich
2017	Assist. Prof. at VU Amsterdam
2021	Assoc. Prof. at VU Amsterdam
2023	Full Prof. at LMU Munich

My Chair at LMU Munich



Relation to SAT Solvers

CDCL-based SAT solvers are generally much more efficient than ground resolution

However, ground resolution can be generalized to nonground resolution, which is competitive

What about the Clause Order?

It is tricky to define precisely
Ask me offline

Intuition behind Variables

Given the function symbols a, b, f ,
the clause

$$C(x)$$

is equivalent to its grounding, the set $\text{ground}(C(x))$:

$$\begin{array}{ll} C(a) & C(b) \\ C(f(a)) & C(f(b)) \\ C(f(f(a))) & C(f(f(b))) \\ \vdots & \vdots \end{array}$$

Nonground Resolution Rule

If the clauses

$$\begin{array}{l} C \vee A \\ \neg A' \vee D \end{array}$$

are contained in $\mathbb{F}$

and substitution σ is the

most general unifier (mgu) of A and A' ,
then add the clause

$$\sigma(C \vee D)$$

to $\mathbb{F}$

Example: Nonground Resolution Rule

Given the clauses

$$C(y) \vee p(a, y)$$

$$\neg p(x, b) \vee D(x)$$

Example: Nonground Resolution Rule

Given the clauses

$$C(y) \vee p(a, y)$$

$$\neg p(x, b) \vee D(x)$$

Example: Nonground Resolution Rule

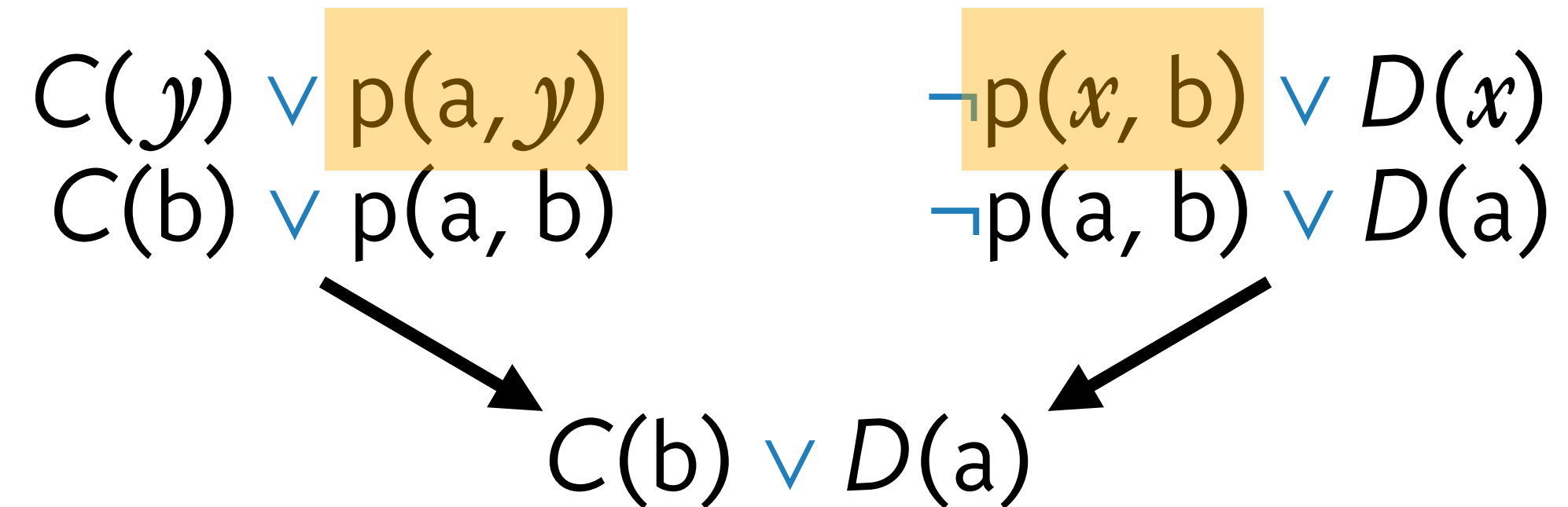
Given the clauses

$$\begin{array}{l} C(y) \vee p(a, y) \\ C(b) \vee p(a, b) \end{array}$$

$$\begin{array}{l} \neg p(x, b) \vee D(x) \\ \neg p(a, b) \vee D(a) \end{array}$$

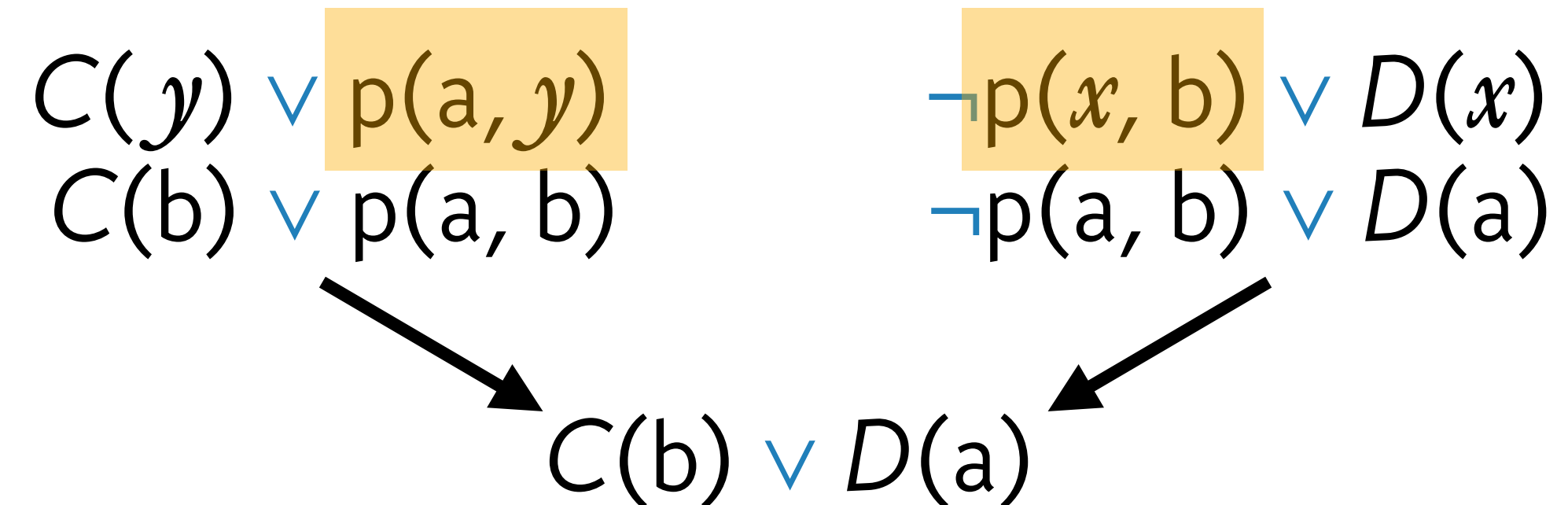
Example: Nonground Resolution Rule

Given the clauses



Example: Nonground Resolution Rule

Given the clauses



1. Unify the p literals, yielding $\sigma = \{x \mapsto a, y \mapsto b\}$
2. Add $\sigma(C(y) \vee D(x))$ to $\mathcal{F}$

Example: A Nonground Resolution Proof

F

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

Example: A Nonground Resolution Proof

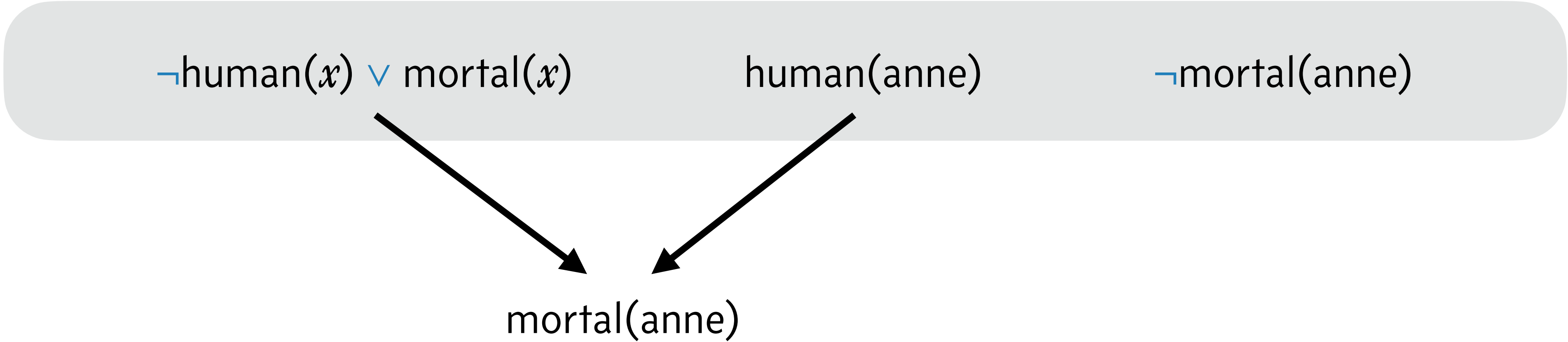
F

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$



Example: A Nonground Resolution Proof

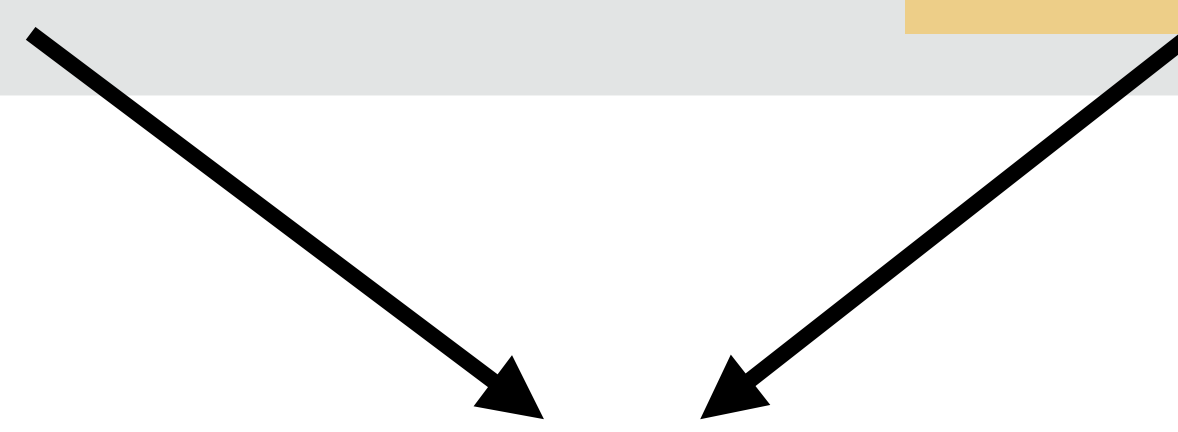
$\mathcal{F}$

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$



Example: A Nonground Resolution Proof

F

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$

Example: A Nonground Resolution Proof

F

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$



Example: A Nonground Resolution Proof

F

$\neg \text{human}(x) \vee \text{mortal}(x)$

$\text{human}(\text{anne})$

$\neg \text{mortal}(\text{anne})$

$\text{mortal}(\text{anne})$



Nonground Redundancy Criterion

A nonground clause C is redundant w.r.t. nonground $\mathcal{F}$
if each clause in $\text{ground}(C)$ is redundant w.r.t. $\text{ground}(\mathcal{F})$

Nonground Redundancy Criterion

A nonground clause C is redundant w.r.t. nonground $\mathcal{F}$ if each clause in $\text{ground}(C)$ is redundant w.r.t. $\text{ground}(\mathcal{F})$

Examples:

- $a(x)$ makes $a(x) \vee b(x)$ redundant
- $b(x) \vee a(x)$ and $\neg a(x) \vee c(x)$ make $b(x) \vee c(x)$ redundant

Soundness of Nonground Resolution

If $\text{ground}(C \vee A)$ and $\text{ground}(\neg A' \vee D)$ are all true,
then each clause in $\text{ground}(\sigma(C \vee D))$ is true

Proof idea: Similar to the ground case

Completeness of Nonground Resolution

If the nonground clause set $\mathcal{F}$ is initially unsatisfiable and inferences are performed fairly, then $\mathcal{F}$ will eventually contain $\perp$

Completeness of Nonground Resolution

If the nonground clause set $\mathcal{F}$ is initially unsatisfiable and inferences are performed fairly, then $\mathcal{F}$ will eventually contain $\perp$

Proof idea:

- By fairness, the limit is saturated
(all inferences have been done or are redundant)
- By static completeness,
the limit contains $\perp$

Static Completeness of Nonground Resolution

If a clause set $\mathcal{F}$ is unsatisfiable and saturated,
then $\mathcal{F}$ contains $\perp$

Static Completeness of Nonground Resolution

If a clause set $\mathcal{F}$ is unsatisfiable and saturated,
then $\mathcal{F}$ contains $\perp$

Proof idea:

- We show that $\text{ground}(\mathcal{F})$ is saturated
(nonground inferences are overapproximation)
- By static completeness of ground resolution,
 $\text{ground}(\mathcal{F})$ contains $\perp$
- Hence $\mathcal{F}$ contains $\perp$

Approximation of Ground Inferences

$\mathbb{F}$

$p(x) \vee q(x)$

$\neg q(y)$

saturated

Approximation of Ground Inferences

F

$p(x) \vee q(x)$

$\neg q(y)$

saturated

$\text{ground}(F)$

$p(a) \vee q(a)$

$p(b) \vee q(b)$

$p(c) \vee q(c)$

$\vdots$

$\neg q(a)$

$\neg q(b)$

$\neg q(c)$

$\vdots$

Approximation of Ground Inferences

F

$p(x) \vee q(x)$

$\neg q(y)$

saturated

$\text{ground}(F)$

$p(a) \vee q(a)$

$p(b) \vee q(b)$

$p(c) \vee q(c)$

$\vdots$

$\neg q(a)$

$\neg q(b)$

$\neg q(c)$

$\vdots$

saturated?

Approximation of Ground Inferences

F

$p(x) \vee q(x)$

$\neg q(y)$

saturated

$\text{ground}(F)$

$p(a) \vee q(a)$

$p(b) \vee q(b)$

$p(c) \vee q(c)$

$\vdots$

$\neg q(a)$

$\neg q(b)$

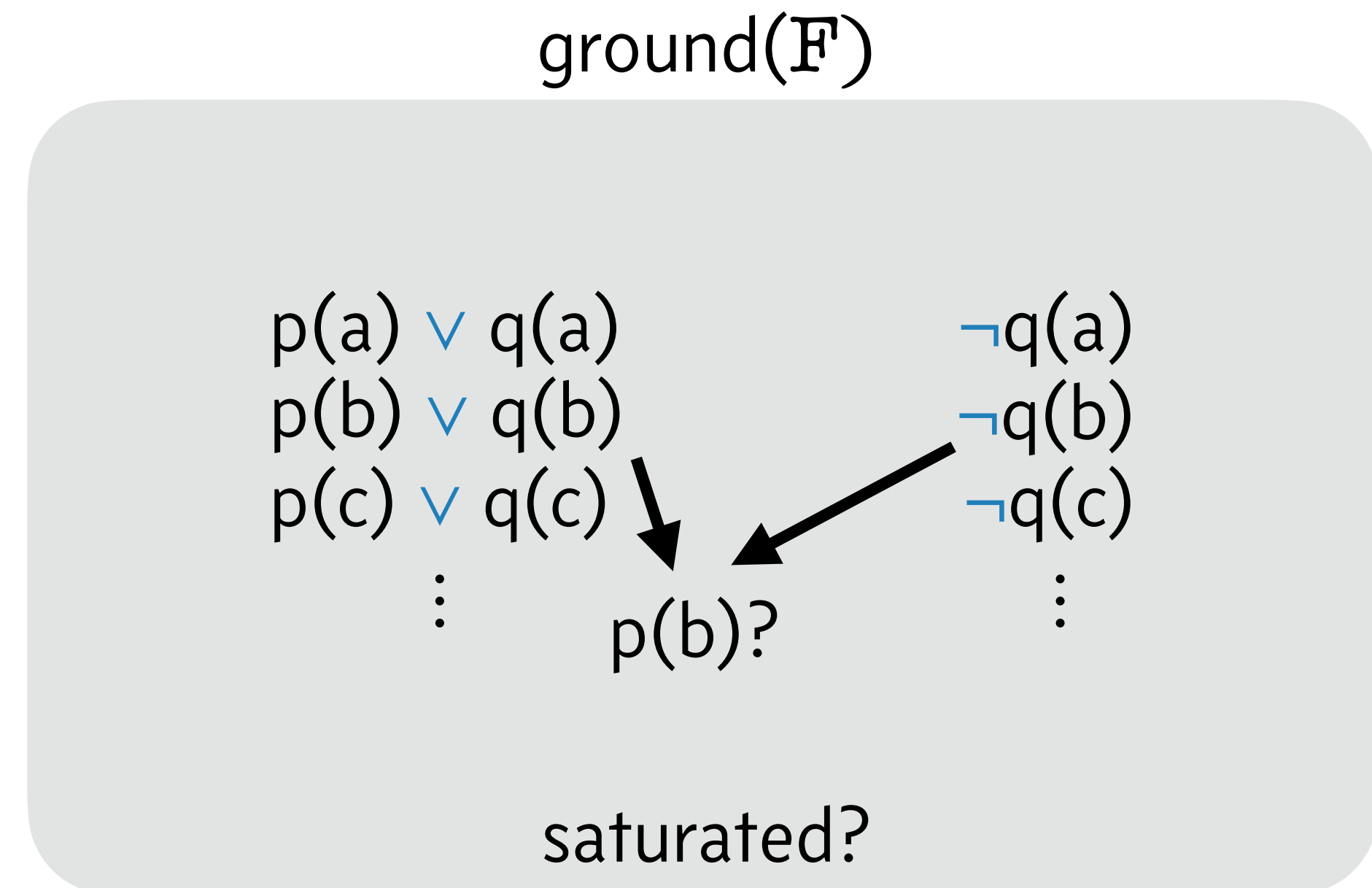
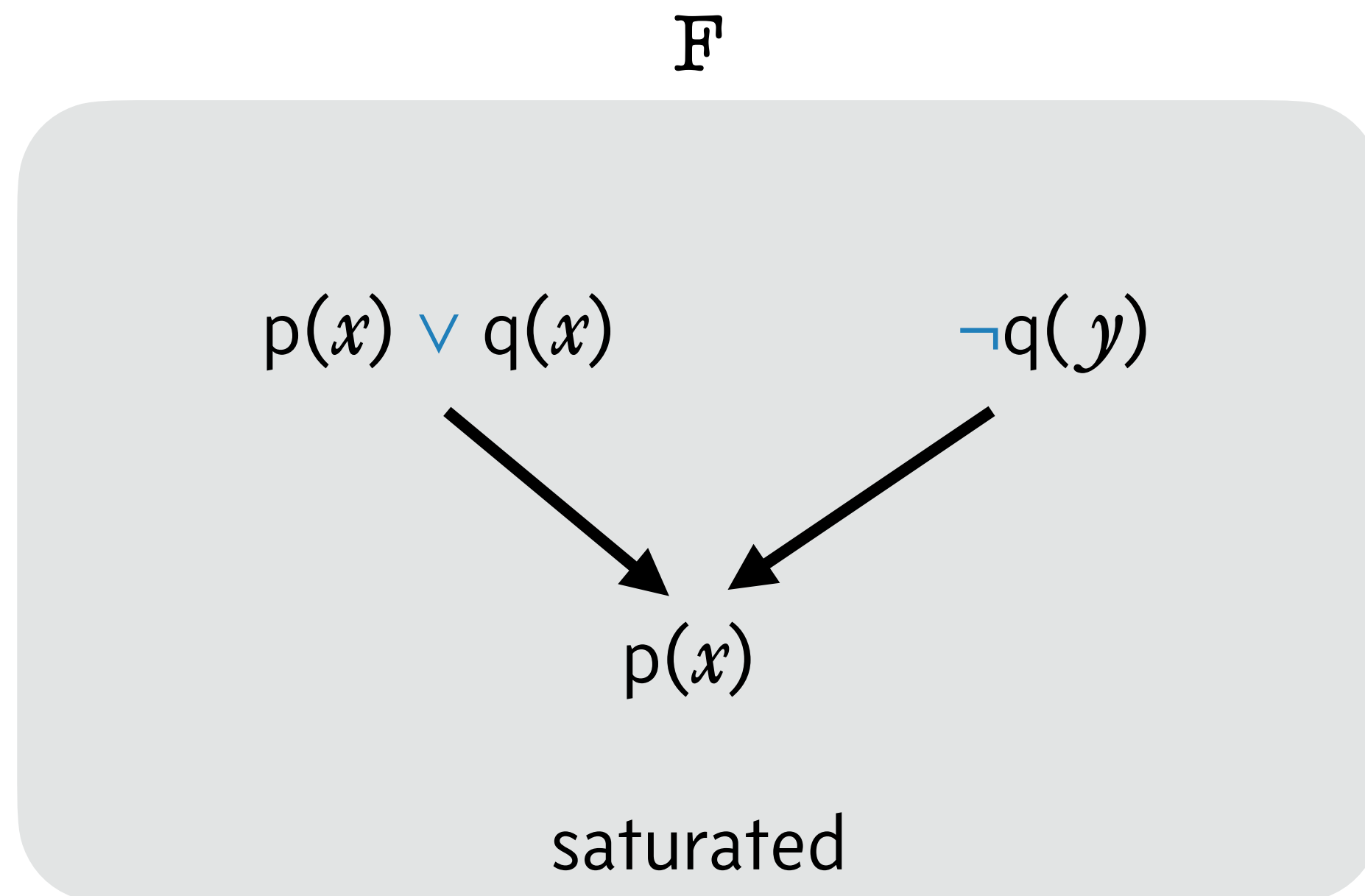
$\neg q(c)$

$\vdots$

$p(b)?$

saturated?

Approximation of Ground Inferences



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A Comprehensive Framework for Saturation Theorem Proving

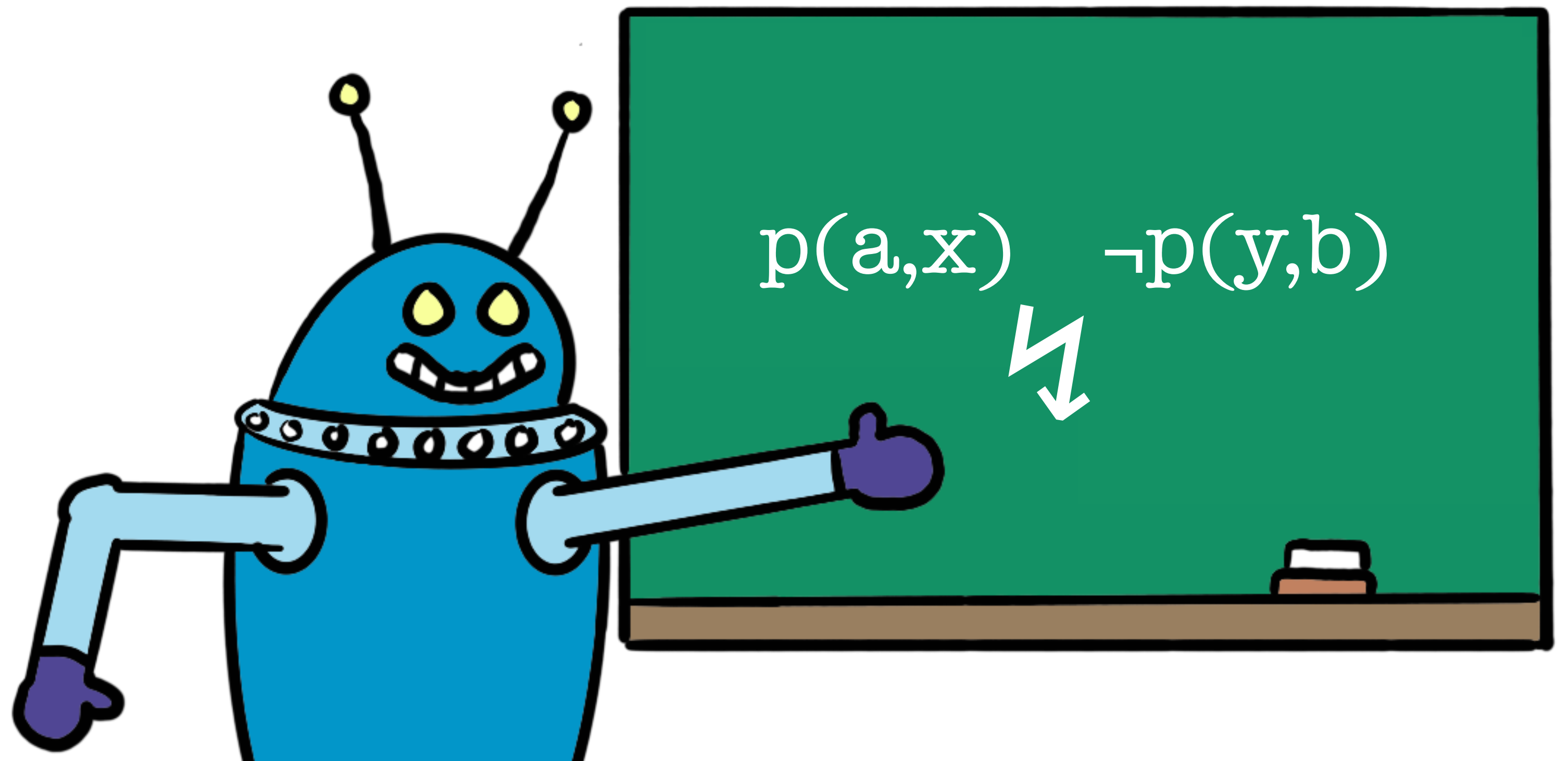
U. Waldmann, S. Turret, S. Robillard, and J. Blanchette
Journal of Automated Reasoning 66(4), 2022

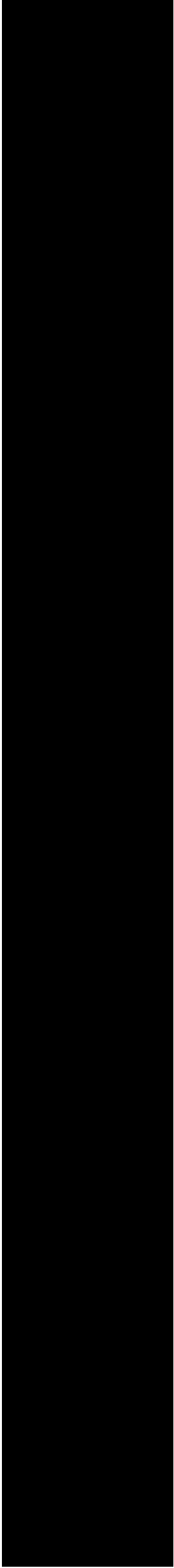
Provers and Solvers

Lecture 1: Resolution

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Blah

